

Original article

A New communication efficient federated online distillation for Non-IID Data: Application to fetal brain ultrasound plane classification

Hassan Salman^{a,*}, Jean-François Pradat-Peyre^b, Sonia Guehis^c, Nour Chrara^d,
Chamseddine Zaki^e, Abbass Nasser^f

^a CNRS – Sorbonne Université, LIP6 UMR, 7606, France

^b CNRS & Paris Nanterre University – Sorbonne Université, LIP6 UMR, 7606, France

^c CNRS & Paris Nanterre University, PSL University, Univ. Paris-Dauphine, LAMSADE UMR 7243, France

^d Computer Science Department, Modern University for Business and Science, Damour, Lebanon

^e College of Engineering and Technology, American University of the Middle East, Egaila, 54200, Kuwait

^f USEK Business School, Holy-Spirit University of Kaslik (USEK), PoBox446, Jounieh, Lebanon



ARTICLE INFO

Keywords:

Federated learning
Knowledge distillation
Attention mechanism
Non-IID data
Communication efficiency
Fetal brain ultrasound

ABSTRACT

Machine learning has achieved notable success in medical imaging; however, its reliability remains constrained by the quality and quantity of training data, particularly in specialized domains such as fetal brain ultrasound. Federated learning (FL) offers a collaborative alternative by enabling multiple institutions to train shared models without exposing sensitive data. Yet, its effectiveness often degrades under non-independent and identically distributed (non-IID) data and its communication demands can be prohibitive. To address these challenges, we propose FedFB, a communication-efficient federated learning framework that integrates online ensemble knowledge distillation for privacy-aware collaborative learning. In FedFB, multiple teacher models trained locally on private datasets collectively distill their knowledge into a lightweight dual-branch student network enhanced with an auxiliary attention–convolution module, without increasing model complexity. The distillation process leverages a small auxiliary public dataset for soft-label transfer, reducing data exposure by avoiding the exchange of raw private data or model parameters, while relying on shared supervisory signals for knowledge transfer. Experimental results on fetal brain ultrasound datasets demonstrate that FedFB achieves 91.07% accuracy, 91.28% precision, 91.03% recall, and 91.15% F1-score under the non-IID conditions, while reducing communication overhead by more than 85% compared to traditional FL methods. Furthermore, the framework's robustness was validated on brain tumor MRI and chest X-ray datasets, confirming its generalization capability across distinct medical imaging modalities.

1. Introduction

Machine learning (ML) and deep learning (DL) have achieved remarkable progress across a wide range of fields, including medical imaging [1]. Yet, their performance remains tightly bound to the quality, quantity, and diversity of the data used for training. In clinical practice, data are often limited, sensitive, and distributed across institutions, making centralized aggregation difficult due to privacy regulations and ethical constraints [2]. These limitations have a profound impact on medical imaging tasks, where the scarcity of labeled data and variability of acquisition protocols can hinder the generalization of even the most advanced DL models [3].

Federated Learning (FL) has emerged as a promising paradigm to address these challenges by enabling decentralized model training across multiple clients or institutions without sharing raw data [4]. Through collaborative optimization, FL improves data utilization while maintaining confidentiality. However, standard FL methods often experience performance degradation when faced with non-independent and identically distributed (non-IID) data and are burdened by high communication costs and rigid model constraints. This motivates the exploration of hybrid strategies that preserve the collaborative benefits of FL while enhancing efficiency and adaptability.

Recently, Knowledge Distillation (KD) has shown strong potential in model compression and transfer learning, allowing lightweight models

* Corresponding author.

E-mail address: Hassan.Salman@lip6.fr (H. Salman).

<https://doi.org/10.1016/j.neuri.2026.100272>

Received 3 December 2025; Received in revised form 22 February 2026; Accepted 13 March 2026

Available online 21 March 2026

2772-5286/© 2026 The Authors. Published by Elsevier Masson SAS. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

to learn from complex teacher networks through soft-label supervision [5]. Combining FL with KD has given rise to Federated Knowledge Distillation (FKD) [6], a communication-efficient alternative that aggregates knowledge from local teacher models via public data rather than exchanging model parameters. As illustrated in Fig. 1, each client produces soft predictions on a shared auxiliary dataset, which are aggregated at a central server to train a global student. This approach minimizes bandwidth requirements, supports heterogeneous model architectures, and enhances robustness across non-IID distributions critical aspects in medical imaging applications.

Within this context, the classification of fetal brain ultrasound images represents a clinically significant, yet data constrained problem. The fetal brain is one of the most intricate and vital organs in early development, governing neurological, motor, and cognitive functions. During pregnancy, its rapid structural evolution requires continuous monitoring, as abnormalities in this period may result in long term developmental disorders [7]. Ultrasound (US) imaging remains the standard modality for fetal brain assessment owing to its safety, accessibility, and real time capabilities [8]. Clinicians routinely analyze three specific anatomical plane transventricular, transthalamic, and transcerebellar which form the basis of the fetal brain standard planes illustrated in Fig. 2 and are essential for detecting abnormalities such as hydrocephalus or intracranial hemorrhage [9]. However, capturing these planes consistently is challenging and heavily operator dependent, underscoring the need for automated, reproducible solutions.

Recent advances in DL have shown remarkable promise in medical imaging tasks, including fetal plane classification, anatomical structure segmentation, and biometric measurements [10], [11]. Several studies have developed CNN-based frameworks for detecting key fetal planes or anatomical regions [12], while others have introduced multi-task learning [13] and segmentation models [14] to improve measurement accuracy. Although these approaches demonstrate promising results, they rely on centralized data storage, raising privacy and regulatory concerns while limiting deployment across multiple healthcare institutions. Moreover, heterogeneous imaging conditions, different

scanners, acquisition settings, and populations further restrict model generalization. These factors make fetal brain ultrasound classification an ideal test case for privacy-aware and communication-efficient collaborative learning.

To address these challenges, we propose FedFB, a FL framework that integrates ensemble KD to enable efficient collaboration under non-IID data distributions. In this framework, multiple teacher models trained independently on local datasets transfer their knowledge to a lightweight dual-branch student model, MobileNetV3Small_AC, which combines parallel attention and convolution modules for enhanced feature representation without added complexity. Rather than exchanging model weights, clients share soft predictions on a small public dataset, drastically reducing communication overhead and preserving privacy. Additionally, Maximum Local Variation-Based Unsharp Masking (MLVUM) [15] is incorporated during preprocessing to improve image clarity, suppress speckle noise, and enhance model discriminability. Together, these components form a unified and efficient FL system suitable for deployment in resource constrained medical environments.

The main contributions of this work are as follows.

- We propose FedFB, a federated KD framework that enhances performance under non-IID data while reducing communication cost by more than 85%.
- The framework is optimized for limited datasets, ensuring stable convergence and high accuracy in data scarce clinical applications.
- We design MobileNetV3Small_AC, a lightweight dual-branch student model that integrated convolutional and attention modules in parallel to enrich feature learning without increasing computational demand.
- We validate the proposed approach on fetal brain ultrasound data and further assess its generalization on brain tumor MRI and chest X-ray datasets demonstrating consistent accuracy and scalability across imaging modalities.

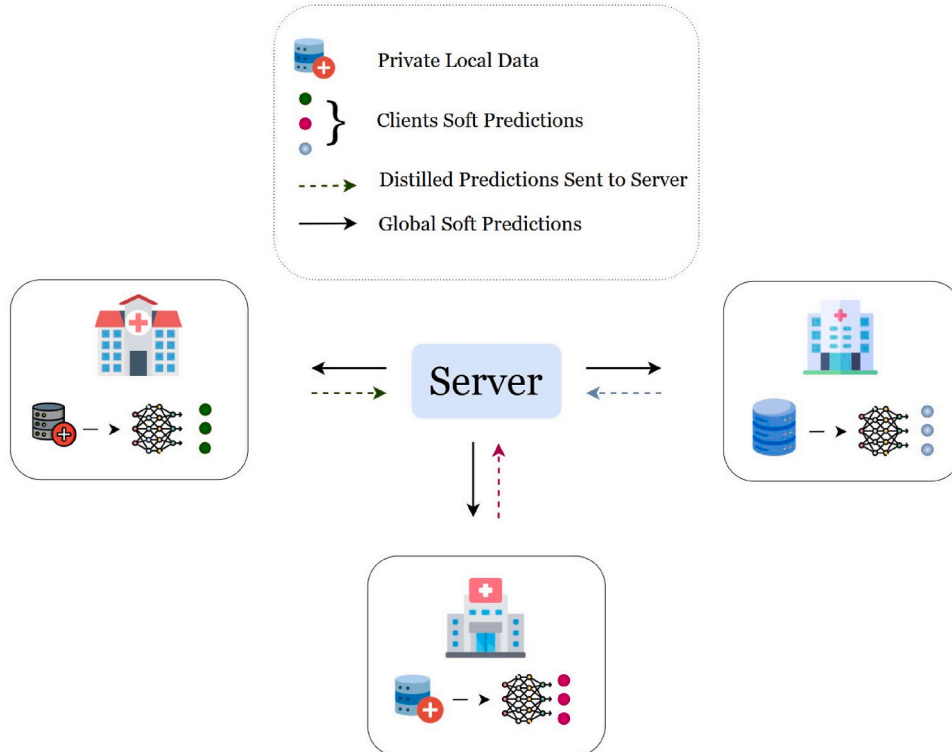


Fig. 1. General Architecture of FKD across distributed clients.

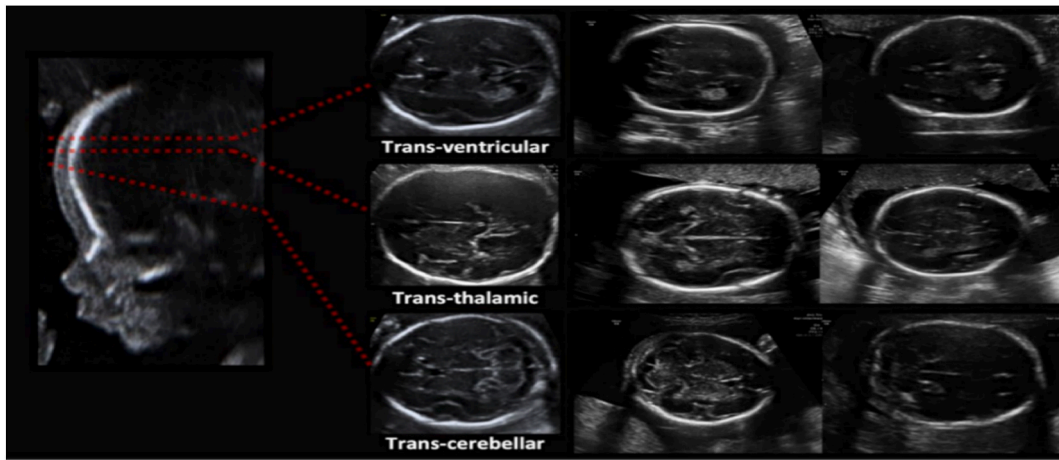


Fig. 2. Standard Planes of fetal brain.

The remainder of this paper is structured as follows: Section 2 discusses the related work. Section 3 describes the proposed FedFB framework, including the model architecture, training strategy and dataset preparation. Section 4 presents the experimental results and comparative evaluation with baseline methods. Section 5 provides a discussion on the framework's adaptability and implementation across other medical imaging datasets. Finally, Section 6 concludes the study and highlights future research directions.

2. Related work

Automated detection of fetal brain standard planes (FBSPs) using DL has gained increasing attention, with numerous studies advancing the state of accuracy, robustness, and efficiency. A CNN-based regression model capable of predicting the six-degree-of-freedom (6D) pose of fetal brain planes achieved localization errors below 3 mm and rotation errors of approximately 6° , demonstrating the feasibility of DL-based navigation and quality control in clinical settings [16]. A more recent pipeline introduced semi-supervised segmentation to extract brain frames from 3D ultrasound volumes and combined them with a pose regression network, improving plane proximity estimation during live scanning [17].

Efforts toward computational efficiency have led to the design of lightweight architectures optimized for real time and resource-limited environments. For instance, LPC-SonoNet integrated light pyramid convolution blocks into the original SonoNet architecture, achieving 97% classification accuracy on a public dataset of 12,400 images across six standard planes (including fetal brain views) while reducing model size to around 4.3 million parameters [18]. Building on this, EfficientNet-based attention-enhanced models further improved parameter efficiency, reporting 96% accuracy top-1 and F1-scores near 0.96 with under one million parameters, while maintaining strong generalization across multiple fetal standard planes [19].

More specialized approaches have focused on fine-grained fetal brain plane classification. One study developed a MobileNetV2-based hybrid branch network that combined parallel convolutional and attention pathways, improving accuracy by up to 7.43% compared with baseline lightweight models [20]. Another utilized U-Net segmentation to isolate the fetal brain region before classification using an ensemble of InceptionV3, ResNet50-V2, and DenseNet-201, reaching 97.68% accuracy and improving interpretability through localized saliency analysis [21]. In parallel, generative modeling approaches such as StyleGAN2-ADA have been explored to synthesize realistic ultrasound planes, demonstrating that combining GAN-generated with augmented data enhances classification of visually similar views like trans-thalamic and trans-ventricular, particularly under limited data availability [22].

While these methods substantially improved efficiency and interpretability, they were developed within centralized training paradigms, limiting scalability and data privacy. KD has been leveraged to transfer performance from complex teacher networks to lightweight students suitable for portable systems. A notable fetal ultrasound method used KD to train a smaller student network that replicated a large teacher's decision behavior, maintaining accuracy while reducing inference cost [23]. Despite these advances, KD-based models remain constrained to centralized settings and cannot support multi-institutional collaboration or privacy preservation [24].

In contrast, FL offers a promising framework for training across distributed data sources without sharing raw images. Early studies have shown its potential for fetal ultrasound, such as a contrastive prototype-based FL method that addressed noisy labels and client imbalance, improving robustness under privacy constraints, although it did not focus on brain-specific planes or employ distillation [17]. FKD expands on this concept by replacing full model weight sharing with soft-label transfer, thereby supporting model heterogeneity, reducing communication overhead, and enhancing generalization in non-IID conditions [6].

Recent research in broader medical imaging further demonstrates the synergy between FL and KD. Approaches like FedISCA have validated FKD scalability across modalities [25], and a one-shot FL framework combining feature guided rectified flow with dual-layer KD achieved efficient aggregation and alignment across clients, outperforming baselines on non-IID datasets [26]. Likewise, large scale frameworks such as UltraFedFM confirmed the feasibility of privacy-preserving federated pretraining on over a million ultrasound images spanning multiple organs and institutions, marking a significant step toward practical clinical deployment [27]. Despite these advances, no prior work has integrated FKD into fetal brain ultrasound plane classification, leaving an important gap addressed by this study.

3. Proposed method

To overcome the challenges posed by the variability of fetal ultrasound images, privacy concerns in clinical data sharing, and the limited generalization of centralized DL models, we introduce FedFB, a communication-efficient FL framework that leverages online ensemble KD. Unlike traditional FL approaches that aggregate model parameters, FedFB facilitates collaboration among clients by allowing each to train its own complex teacher model locally and contribute only soft-label predictions generated on a shared auxiliary dataset.

The FedFB framework is structured to operate across multiple sites, simulating real-world clinical scenarios where data heterogeneity and limited bandwidth are common. Each client has access to a private

dataset and trains a local teacher model without exposing patient data or model parameters. These teacher models are used to produce soft predictions on a publicly shared auxiliary dataset, which is transmitted to a central server. These soft-label predictions contain only aggregated class probabilities rather than raw image features, substantially reducing data exposure compared to parameter sharing, although without providing formal privacy guarantees [28]. A lightweight student model integrating a convolutional and attention branch is trained using aggregated knowledge from the ensemble of teachers. This process not only reduces communication cost but also mitigates the performance drop typically associated with non-IID data distributions across clients.

FedFB integrates multiple components tailored to the challenges of fetal brain ultrasound plane classification.

- A preprocessing pipeline that enhances image quality using MLVUM and increases variability through targeted data augmentation.
- Client-specific teacher models trained locally on private data.
- A shared auxiliary dataset for producing soft labels during online distillation.
- A central lightweight student model (MobileNetV3Small_AC) with a dual-branch design to efficiently learn from aggregated knowledge.

The complete workflow is illustrated in Fig. 3, showing distributed teacher training, soft-label aggregation, and centralized student distillation. The following subsections describe each component of FedFB in detail.

3.1. Teacher model

In our FedFB framework, we adopt the MobileNetV3-Large architecture as the backbone of the teacher model. This choice is motivated by the architecture's balance of computational efficiency and representational power, which is essential in federated environments constrained by memory and communication bandwidth. Importantly, we utilize the vanilla MobileNetV3-Large variant providing a streamlined and lightweight design suitable for scalable federated deployment. MobileNetV3-Large is a convolutional neural network optimized for mobile and embedded vision tasks. It builds upon the ideas from MobileNetV1 and V2 by integrating inverted residual blocks, squeeze-and-excitation (SE)

modules, and the h-swish activation function [29]. However, in our configuration, all SE modules are retained. Fig. 4 shows a simplified diagram illustrating the modified MobileNetV3-Large teacher model used in our framework.

The teacher model's classifier head replaces the original MobileNetV3 classification block, enhancing it with deeper layers and stronger regularization. This contributes to a more stable and robust teacher signal during distillation, particularly when working with noisy or imbalanced ultrasound datasets. To expedite convergence and avoid overfitting on limited client data, we freeze the backbone (feature extractor) parameters during training. Only the custom classifier head is fine-tuned locally on each client. This reduces the number of trainable parameters significantly, allowing faster training while preserving the rich features learned from ImageNet.

Table 1 below summarizes the architectural configuration and parameter count of the MobileNetV3-Large teacher model used in FedFB.

3.2. Student model

To address the efficiency needs of FL while still maintaining strong classification performance, we designed a student model that builds on the MobileNetV3-Small backbone. Rather than increasing network depth which often adds training time and computational cost, we opted for a parallel structure that enhances feature richness without inflating model size. This dual-branch design allows the network to process the same features in two distinct ways: one stream focuses on learning where to attend in the feature space, while the other enhances spatial understanding through additional convolutions. The outcome is a compact, low overhead model that converges quickly and generalizes well in our decentralized training setup.

As shown in Fig. 5, the student model takes an input image and first passes it through the pretrained backbone for core feature extraction. From there, the extracted features are fed into two independent modules: an attention path that learns to emphasize the most informative regions, and a convolutional path that refines spatial details. These two streams are then combined and fused into a unified representation before entering the final classifier. The fusion of the attention and convolution branches is performed through channel-wise concatenation

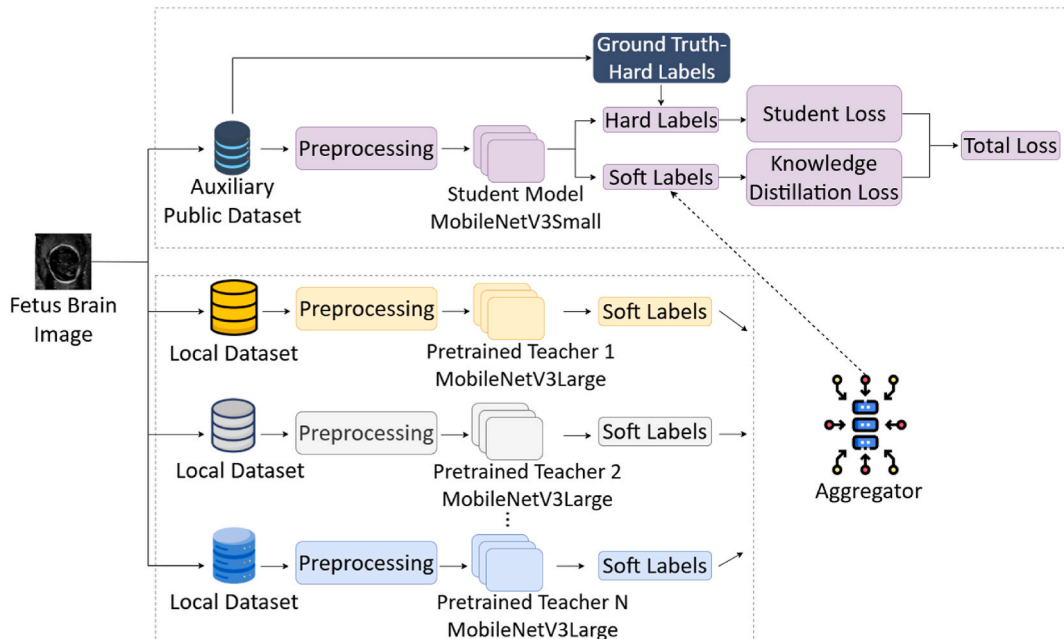


Fig. 3. FedFB high-level architecture: Clients train local teacher models on private data and share only logits on a public dataset. A server-side ensemble aggregates these logits to guide the training of a global student model via KD.

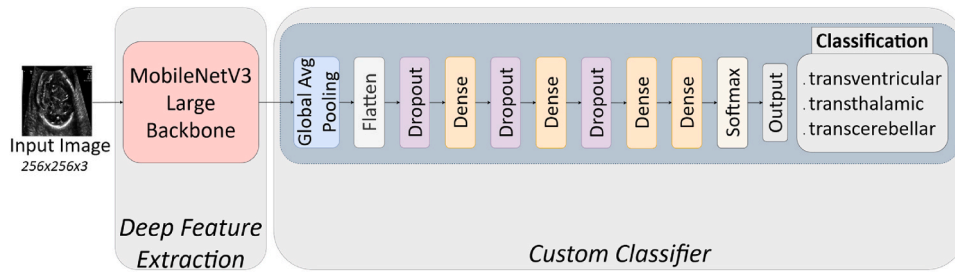


Fig. 4. Teacher model architecture.

Table 1
Architecture of Teacher model.

Network	Layer (type)	Kernels, Activation	Output Shape	#Param
Pre-trained	Base model = MobileNetV3Large	–	(None, 8, 8, 960)	2,971,952
Classification	Global Avg. Pool	–	(None, 960)	0
	Dense	960 → 512, ReLU	(None, 512)	492,032
	Dropout	p = 0.2	(None, 512)	0
	Dense	512 → 128, ReLU	(None, 128)	65,664
	Dropout	p = 0.2	(None, 128)	0
	Dense	128 → 3, Softmax	(None, 3)	387
Total Trainable				558,083
Frozen				2,971,952
Total Parameters				3,530,035

followed by adaptive feature compression. Specifically, after extracting features from the MobileNetV3-Small backbone, the attention branch processes these features through an AttentionBlock that applies element-wise multiplication to highlight salient anatomical regions, while the convolution branch applies additional convolutional transformations via a ConvBlock to extract complementary spatial features. The attention-refined features and the convolution-extracted features are then concatenated along the channel dimension, resulting in a combined feature tensor. This concatenated representation is subsequently processed by a convolution layer followed by adaptive average pooling and fully connected layers for final classification. This concatenation-based fusion strategy allows the model to jointly leverage both attention-weighted spatial information and additional learned convolutional features, providing a richer and more discriminative feature representation than either branch alone. The subsequent dimensionality reduction via convolution ensures computational efficiency while maintaining expressive power. The diagram illustrates how each module is modularly integrated and how the architecture avoids unnecessary stacking by making better use of feature diversity.

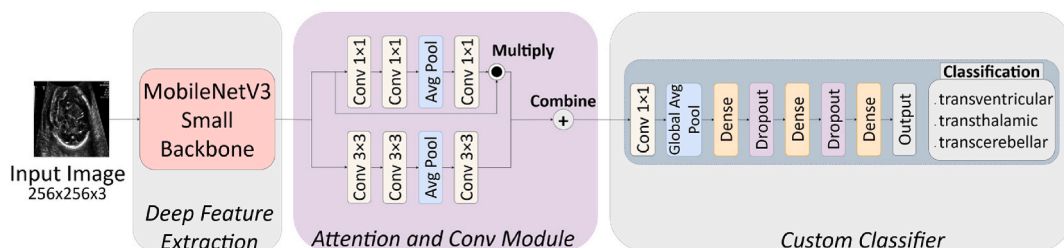


Fig. 5. Illustration of the student model architecture.

Further architectural specifics are listed in Table 2, including the dimensional transitions, activation choices, and parameter counts for each block. The table highlights how the design maintains a total parameter count below 2.9 million, making it well suited for federated contexts where computation and memory are limited. The classifier follows a simple but effective layout with dropout regularized dense layers leading to a final three class output. Overall, this architecture is a practical middle ground: expressive enough to capture meaningful visual patterns, and light enough for real-world deployment. The design aligns with findings from prior work suggesting that multi-branch feature encoding can outperform deeper sequential models, especially in constrained settings [30].

3.3. Dataset

This study utilizes a curated subset of a publicly available ultrasound dataset originally compiled by Burgos-Artizzu et al. [31], which includes images of various maternal and fetal anatomical regions. For our purposes, we focus exclusively on fetal brain images, specifically those categorized under three essential neurosonographic planes: trans-thalamic, trans-cerebellum, and trans-ventricular. From the full dataset, we extract 2950 labeled brain images with an even distribution across these three classes.

Despite this balance, the dataset remains relatively small in the context of DL particularly within FL, where data is distributed across multiple clients. Each client receives only a fragment of the dataset, amplifying challenges such as limited sample availability, non-IID data distributions, and potential overfitting. To help address these issues and improve generalization, we apply standard augmentation techniques exclusively to the training set pre-processing, such as random rotation and horizontal flipping. These augmentations increase the diversity of the data and expand the total number of effective images to 4770.

An overview of the data split is illustrated in Fig. 6, confirming that each class remains balanced across the training, validation, and test sets with 1136 samples per class for training, 80 for validation, and 374 for testing. The publicly available dataset used in this study, accessible at (<https://zenodo.org/records/3904280>), provides a predefined training-validation-test split. In this work, we adopted the official split used in prior studies to maintain consistency and enable fair comparison across methods.

Table 2
Detailed layer configuration of the student model.

Network parts	Layer (type)	Kernels, Activation	Output Shape	#Param
Pre-trained	Base model = MobileNetV3Small	–	(None, 8, 8, 576)	927,008
Branch-structure	Attention-module	Conv1 × 1, ReLU	(None, 8, 8, 128)	73,856
		Conv1 × 1, ReLU	(None, 8, 8, 32)	4128
		Avg-Pool	(None, 8, 8, 32)	0
		Conv1 × 1, ReLU	(None, 8, 8, 1)	33
	Convolution-module	Multiply (Attention × Features)	(None, 8, 8, 576)	0
		Conv3 × 3, ReLU	(None, 8, 8, 128)	663,680
		Conv3 × 3, ReLU	(None, 8, 8, 64)	73,792
		Max-Pool	(None, 4, 4, 64)	0
		Conv3 × 3, ReLU	(None, 4, 4, 32)	18,464
		–	(None, 8, 8, 608)	0
Merge	Concatenate (576 + 32)	–	(None, 8, 8, 608)	0
Classification	Conv1 × 1, ReLU	608 → 256	(None, 8, 8, 256)	155,904
	Global Avg. Pool	–	(None, 256)	0
	Dense	256 → 128, ReLU	(None, 128)	32,896
	Dropout	p = 0.2	(None, 128)	0
	Dense	256 → 64, ReLU	(None, 64)	8256
	Dropout	p = 0.1	(None, 64)	0
	Dense	64 → 3, Softmax	(None, 3)	195
Total Trainable				1,031,204
Frozen				927,008
Total Parameters				1,958,212

3.3.1. Pre-processing

Ultrasound images often suffer from limited visual clarity, largely due to the presence of speckle noise and inherently low contrast [15]. This noise manifests as a fine, grainy pattern that disrupts the image quality. It is primarily caused by the interference of returning echoes

scattered from microscopic structures, which are frequently smaller than the imaging system's spatial resolution. In addition to this noise, poor contrast between soft tissues further complicates the visibility of anatomical details [32], making accurate identification and interpretation of key brain structures particularly challenging in fetal imaging. To ensure the quality and consistency of fetal brain ultrasound images across clients, we adopt a two-stage pre-processing pipeline consisting of image enhancement and data augmentation. First, we adopt Maximum Local Variation-Based Unsharp Masking (MLVUM) to improve visual clarity and suppress speckle noise. MLVUM computes a local variation map for each image based on intensity gradients:

$$L(x, y) = \max_{(i, j) \in \Omega} |I(x, y) - I(x + i, y + j)| \quad (1)$$

Where Ω is a local neighborhood window and $I(x, y)$ is the pixel intensity at location (x, y) .

This variation map guides selective sharpening using:

$$S(x, y) = I(x, y) + \lambda \cdot (I(x, y) - G(x, y)) \cdot M(x, y) \quad (2)$$

where $I(x, y)$ is the original image, $G(x, y)$ is the Gaussian-blurred version, λ is the sharpening coefficient, and $M(x, y)$ is a binary mask derived from thresholded local variations. This enhances critical anatomical structures such as the lateral ventricles and thalami across the transventricular, transthalamic, and transcerebellar planes. The results of applying the MLVUM method to the fetal brain US images are presented in Fig. 7. The image distinctly demonstrates enhanced edges, producing a sharper overall appearance while effectively minimizing the presence of speckle noise.

Following enhancement, we apply a set of data transformations to increase variability and promote generalization. To ensure unbiased model evaluation, augmentation is strictly limited to training images, while validation and test sets receive only deterministic pre-processing. Specifically, we apply random rotations within $\pm 40^\circ$ and resize images to 256×256 resolution. These transformations simulate real-world scanning variability and help the model learn robust representations across diverse clinical conditions. Together, MLVUM and augmentation ensure that both teacher and student models are exposed to high-quality, varied training inputs that are consistent with ultrasound imaging scenarios.

3.3.2. Data partitioning and auxiliary dataset

To simulate a realistic FL environment, we partition the training dataset across multiple clients using both IID and non-IID schemes. Before applying client partitioning, we first extract a portion of the training set to create a disjoint auxiliary public dataset, ensuring methodological rigor in the KD process.

Auxiliary Dataset Construction: We randomly sample a proportion ρ of the complete training set to create the auxiliary public dataset D_{aux} :

$$|D_{aux}| = \rho \cdot |D_{train}| \quad (3)$$

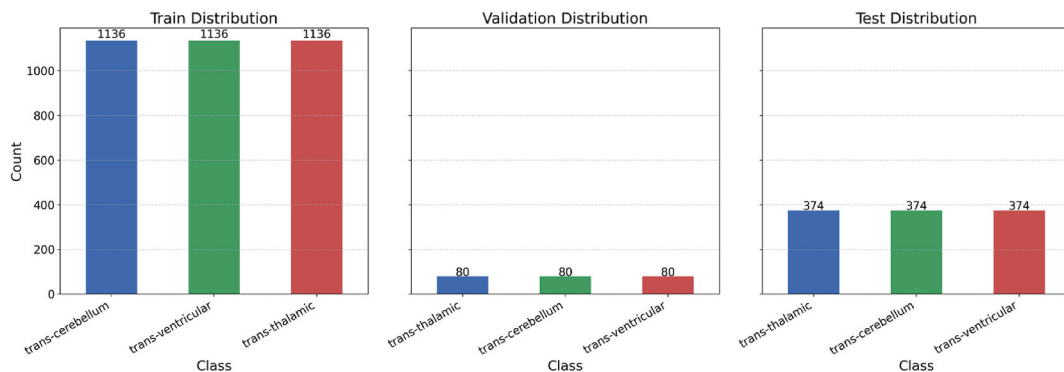


Fig. 6. Dataset distribution.

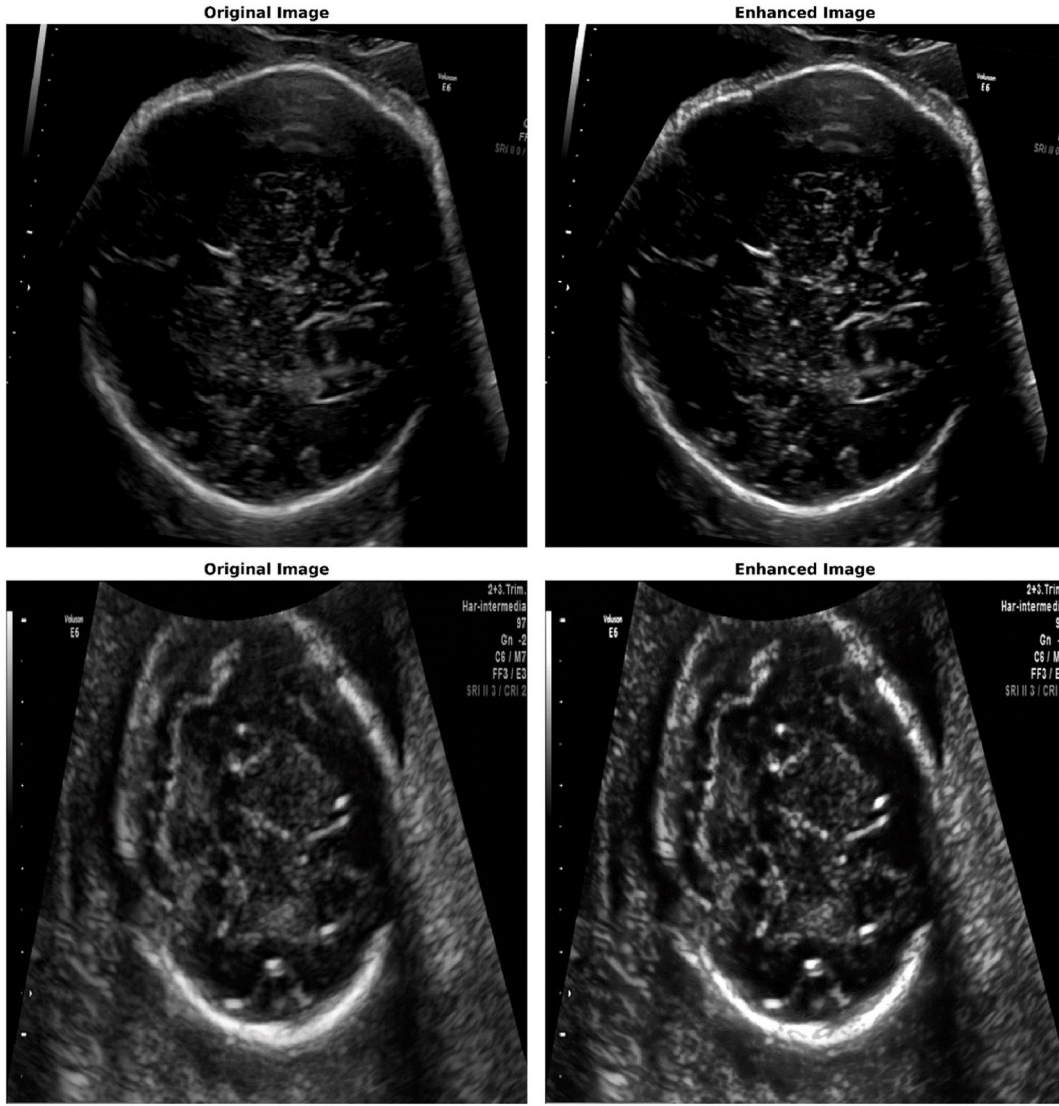


Fig. 7. Enhanced US fetal brain images with MLVUM.

where ρ is set to 0.4 in our experiments. The auxiliary dataset retains ground-truth labels from the original training annotations, enabling dual supervision during student training through both hard labels and aggregated soft teacher predictions. Critically, the auxiliary dataset is extracted exclusively from the training split, while validation and test sets remain completely separate and are never used for KD. After extracting D_{aux} , the remaining portion of the training set forms a disjoint client pool D_{client} used exclusively for partitioning among clients:

$$|D_{client}| = (1 - \rho) \cdot |D_{train}| \quad (4)$$

This disjoint splitting ensures complete separation between the auxiliary dataset and all client training data, preventing teachers from generating predictions on images they encountered during local training. The auxiliary dataset serves solely as a neutral input source on which all client teachers produce soft labels. These soft predictions are then aggregated at the server to guide the student model, maintaining client privacy throughout the distillation process.

For client data distribution from D_{client} , we apply IID and non-IID partitioning schemes. In the IID setup, the client pool is randomly and evenly distributed across clients, ensuring that each client sees an approximately similar class distribution. For the non-IID scenario, we implement Dirichlet-based sampling to introduce data heterogeneity across clients, which better reflects practical deployment settings. Spe-

cifically, we sample client data distributions from a Dirichlet distribution:

$$p^{(c)} \sim \text{Dir}(\alpha), \forall c \in \{1, 2, 3\} \quad (5)$$

where $p^{(c)} = [p_1^{(c)}, p_2^{(c)}, \dots, p_k^{(c)}]$ represents the proportion of class c assigned to each of the K clients, and α controls the level of heterogeneity. Lower values of α result in more skewed distributions, leading to stronger non-IID conditions.

This approach allows us to maintain client privacy while still enabling effective global learning, despite the absence of external public data and the overall limited size of the fetal brain dataset.

3.4. Training Models

Our training strategy combines localized supervision with centralized knowledge aggregation, leveraging a teacher-student paradigm designed for FL. Clients independently train teacher models on their local data, while a global student model is trained concurrently through online KD. This means that instead of waiting for all teachers to finish training, the student is updated continuously across rounds using the soft predictions of the teachers in each iteration.

3.4.1. Client-side teacher training

Each client receives a subset of the training data in both IID or non-IID settings depending on the experimental setup and independently fine-tunes a MobileNetV3-Large model as its local teacher. To avoid overfitting and to reduce computational overhead, we freeze the backbone of the model and train only the classifier head. This setup ensures that the rich pretrained features from ImageNet are preserved while allowing the model to adapt to fetal brain plane classification.

Local training is supervised using the standard categorical cross-entropy loss on the client's private labeled dataset. For a batch of training examples $\{(x_i, y_i)\}_{i=1}^N$, the client's loss function is defined as:

$$\mathcal{L}_{client}(\hat{y}, y) = -\log \sum_{c=1}^C y_c \cdot \text{softmax}(W_c \cdot \hat{y} + b_c) \quad (6)$$

where C is the number of classes, $\hat{y}_i^{(c)}$ is the predicted logit for class c , and y denotes the true label for class c . W_c and b_c are the learnable weights and biases of the classifier head. This loss encourages each teacher to produce confident predictions aligned with the client's local distribution.

Clients train their teacher models independently, without exchanging model parameters or raw data. As a result, each teacher model inherently adapts to the specific data distribution and class imbalance present within its local subset. These distinct, locally adapted representations are later aggregated during the distillation phase, allowing the student model to learn from a more diverse representation of the task.

3.4.2. Global-side student training

The student model in our framework is trained through a centralized yet privacy-aware process using online KD. Unlike traditional approaches where teacher models are fully trained beforehand and distillation occurs afterward (offline), our method performs online distillation, meaning the student is updated iteratively during the same rounds in which the teachers generate predictions. In each round, teachers first undergo training epochs on their local data to refine their knowledge, then generate predictions on the auxiliary public dataset, which are aggregated to train the student for several epochs. This design reduces training time and enables the student to benefit from the evolving knowledge of teacher models across rounds.

The student architecture, as described in Section 3.2, is based on a lightweight MobileNetV3 model that combines a frozen MobileNetV3-Small backbone with a parallel Attention-Convolution (AC) feature fusion module and a custom classifier head. This structure is chosen to ensure both representational richness and computational efficiency, which is a key requirement in federated setups where the student may be deployed across diverse devices.

During each round, the public auxiliary data D_{aux} is distributed to all clients and processed in mini-batch for inference. Each client uses its locally trained teacher model to produce soft predictions over these shared samples. These logits are then sent to the central aggregator, which computes the average logits across all clients for each data point as described below:

$$\bar{z}_i = \frac{1}{K} \sum_{k=1}^K z_i^{(k)} \quad (7)$$

where $z_i^{(k)}$ represents the teacher logits from client k for sample x_i , and K is the total number of participating clients.

The student is then trained on this aggregated supervision using a hybrid loss that combines hard supervision from ground truth labels with soft supervision from the averaged teacher predictions. The total loss function is given by:

$$\mathcal{L}_{student} = \alpha \cdot \mathcal{L}_{CE}(y, \hat{y}_s) + (1 - \alpha) \cdot T^2 \cdot \mathcal{L}_{KL}\left(\sigma\left(\frac{\bar{z}}{T}\right), \sigma\left(\frac{\hat{y}_s}{T}\right)\right) \quad (8)$$

Where $\hat{y}_s$ are the student logits, $\bar{z}$ are the aggregated teacher logits, $\sigma(\cdot)$ is the SoftMax function, T is the distillation temperature, and α balances the hard and soft components of the loss.

The KL divergence encourages the student to imitate the teachers softened outputs, capturing subtle inter class relationships, while the cross-entropy term grounds the model in the true class labels. This hybrid supervision strategy helps the student generalize better, especially under non-IID client conditions where knowledge diversity is high.

At the end of training, the student model encapsulates the combined knowledge of all local teachers, while remaining lightweight and fully trainable without direct access to any private client data. This makes it ideal for centralized deployment or redistribution to edge clients for efficient inference.

The complete training process for client and teacher models are summarized in Algorithm 1.

Algorithm 1: Training Models

Inputs: Full dataset: $D_{train} = \{(x_i, y_i)\}_i^N$, Auxiliary public dataset: $D_{aux} \subset D_{train}$, Local private dataset: $D_k \subset D_{train} \setminus D_{aux}$, Weighting factor for CE & KL loss: α , Distillation Temperature: T , Number of clients: K , learning rate: η , Teacher epochs: E_t , Student epochs: E_s .

Output: Global Student Model M_s

1. Client Initialization
 - 1.1. For each client C_k initialize teacher model M_k with pretrained weights
 - 1.2. Partition $D_{train} \setminus D_{aux}$ across clients using Dirichlet (a_{dir}) or IID scheme
 - Teacher models persist across rounds.
2. Global Student Initialization
 - Initialize student model M_s
 - The student model persists and is updated across rounds.
3. Federated Rounds (Online Distillation)

For round = 1 to R :

 - 3.1. Each Client C_k trains M_k for E_t epochs on local data D_k
 - 3.2. Each client C_k computes logits $z_i^{(k)}$ on D_{aux} using the updated M_k
 - 3.3. Server aggregate logits $\bar{z}_i$
 - 3.4. Server trains student model M_s for E_s epochs on D_{aux} using hybrid loss
 - 3.5. Update M_s via gradient descent with learning rate η
4. Output final student model M_s

4. Experimental results

This section presents the experimental evaluation of the proposed FedFB method. Our goal is to assess how well the student model learns from distributed teacher models in both IID and non-IID settings, and to examine its effectiveness in classifying standard fetal brain planes from ultrasound images. We begin by outlining the experimental configurations, Model evaluation metrics, and classification results used throughout the study. The subsequent sections detail quantitative and qualitative results across different baselines methods. FedFB is compared against well-known baseline FL methods, including FedAvg [33], FedProx [34], and FedDF [35], as well as with a centralized student model used as an additional baseline, to provide a fair and comprehensive performance analysis.

4.1. Setups

All experiments were carried out using PyTorch 1.13.1, executed on a workstation equipped with an NVIDIA Quadro RTX 6000 GPU, CUDA 11.7, and a Python 3.10 environment. The ultrasound dataset used in this study comprises 4770 fetal brain images, divided into three anatomical classes as mentioned in section 3.3. To simulate FL, we evaluate the framework under different client configurations, using 2, 5, and 10 clients. In each case, 60% of the dataset is partitioned among the clients, while the remaining 40% is reserved as a public auxiliary dataset to support online KD.

Data distribution across clients is performed under both IID and non-IID settings. For the non-IID case, a Dirichlet distribution with a con-

centration parameter $\alpha = [0.1, 0.5]$ is used to induce varying class imbalances among clients. Each client trains a local model, initialized with pretrained ImageNet weights. The backbone of the model is frozen, and only the classification head is fine-tuned using categorical cross-entropy loss. Training on each client is done using the Adam optimizer with a learning rate of 1×10^{-4} , weight decay of 5×10^{-4} , and batch size of 32. A ReduceLROnPlateau learning rate scheduler is applied to adaptively decrease the learning rate when validation performance stops improving. The global student model is trained for 10 communication rounds via online KD. In each round, teachers first undergo 5 training epochs on their local data, then generate soft predictions on the auxiliary dataset. The student is then trained for 5 epochs using both the ground truth labels and the aggregated soft outputs from all teachers, resulting in 50 total student training epochs across the 10 rounds. Throughout all experiments, the distillation temperature is fixed at $T = 4$, and the KD weight is fixed at $\alpha = 0.7$ across all communication rounds.

Additional implementation details, including pseudocode for pre-processing, hyperparameter search ranges, and the exact Dirichlet-based partitioning procedure, are provided in the Supplementary Material to ensure full reproducibility.

4.2. Model training protocol

To ensure fair comparison across all methods, we maintain consistent training protocols with explicit control over trainable versus frozen parameters. As described in sections 3.1 and 3.2, both teacher and student models employ a transfer learning strategy where ImageNet-pretrained backbones are frozen while only task-specific layers are trained. Specifically, teachers freeze the MobileNetV3-Large backbone and train only the classifier head, while students freeze the MobileNetV3-Small backbone and train the AC auxiliary module and classifier. Critically, all baseline methods and the centralized baseline use the identical dual-branch MobileNetV3-Small architecture with exactly the same frozen/trainable configuration as our student model frozen backbone parameters and trainable parameters. This ensures that all methods optimize the same number of trainable parameters under identical architectural constraints, with performance differences attributable solely to the FL algorithm rather than parameter count or model capacity disparities.

4.3. Evaluation metrics

To assess the performance of both teacher and student models, we adopt a set of standard classification metrics widely used in multi-class image recognition tasks. These include accuracy, precision, recall, and F1-score, which provide complementary views on model correctness, class-wise performance, and generalization. In addition, we report the communication cost incurred during the federated distillation process, which reflects the practical scalability of the framework.

The above evaluation metrics are defined in the following manner:

$$\text{Accuracy} = \frac{TP_c + TN_c}{\text{Total number of test samples}} \quad (9)$$

$$\text{Precision} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c} \quad (10)$$

$$\text{Recall} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c} \quad (11)$$

$$\text{F1-Score} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (12)$$

where P_c , FP_c , FN_c and TN_c denote the number of true positives, false positives, false negatives, and true negatives, respectively, for class $c \in$

$\{1, 2, 3\}$, and C is the total number of classes.

In federated settings, communication overhead is a critical performance factor. In our FedFB setup, soft predictions are exchanged between clients and the server during each round, making the communication cost considerably lower than traditional weight sharing methods.

The total communication cost across all rounds is computed as shown:

$$\text{Comm}_{\text{total}} = R \cdot K \cdot N \cdot d \quad (13)$$

Where R denotes the number of rounds, K is number of clients, N indicated the size of auxiliary public data per round and d is the dimension of output logits.

4.4. Classification results

During the training process of the student network in different federation sizes and data distributions, the accuracy curves show steady improvement along the 50 global epochs as shown in Fig. 8. The student is trained for 10 communication rounds with 5 training epochs per round, yielding 50 total training epochs. In IID setups with 2, 5, and 10 clients, there was a quite fast gain during the first 10-15 epochs, followed by a slower but stable increase until convergence. For the largest federation with 10 clients, the accuracy started around 33% and passed 92% by the final epoch. In Non-IID settings, especially with $\alpha = 0.1$ where the class imbalance is high, the early learning stage was slower but still managed to reach around 89% at the end. Using the AC module in the student network helped to speed up convergence and make it more stable, probably because it let the model focus more on important brain structures like the thalami and cerebellum. Without AC, the accuracy was more unstable and the final values lower, showing that the model learned less reliable features.

The generalization capability, evaluated by test accuracy across communication rounds, is presented in Fig. 9 for IID settings and Fig. 10 for Non-IID scenarios. In IID configurations, FedFB attained final accuracies of 90.55%, 90.19%, and 91.17% for 2, 5, and 10 clients, respectively, indicating that increasing the federation size did not compromise performance despite fully disjoint local datasets. In Non-IID settings with $\alpha = 0.1$ and $\alpha = 0.5$, initial accuracy was affected by label skew, but by round 10 the models exceeded 89%. The results show that using the soft-label transfer, which is supported by a shared public dataset, can reduce the differences between the data distributions of different clients. This is very important in real practice, since ultrasound scanning protocols and image settings can change a lot from one clinic to another.

The benefit of the AC module is further quantified in Fig. 11. In the 10-client IID scenario, the presence of AC elevated accuracy from 74% to 91.17%, with similar gains observed for the 2- and 5-client settings. Notably, the advantage of AC was most apparent in the early communication rounds. During these initial stages, the student model is trained with limited and heterogeneous supervisory signals aggregated from distributed teachers, making feature learning more sensitive to noise and background variability. The attention branch enables the student to rapidly focus on anatomically relevant regions, while the convolution branch reinforces local spatial structure, leading to more stable and discriminative feature representations. As communication rounds progress and the aggregated supervision becomes increasingly consistent, the relative contribution of the AC module diminishes, resulting in a natural convergence of performance across later rounds. This behavior highlights the role of AC in accelerating early learning, particularly under challenging conditions such as fetal motion or suboptimal probe positioning.

In addition to accuracy, FedFB maintained high classification quality across precision, recall, and F1-score metrics. As shown in Fig. 12, the best performing 10-client IID configuration achieved 0.912 precision, 0.911 recall, and 0.901 F1-score. Even in the most challenging non-IID

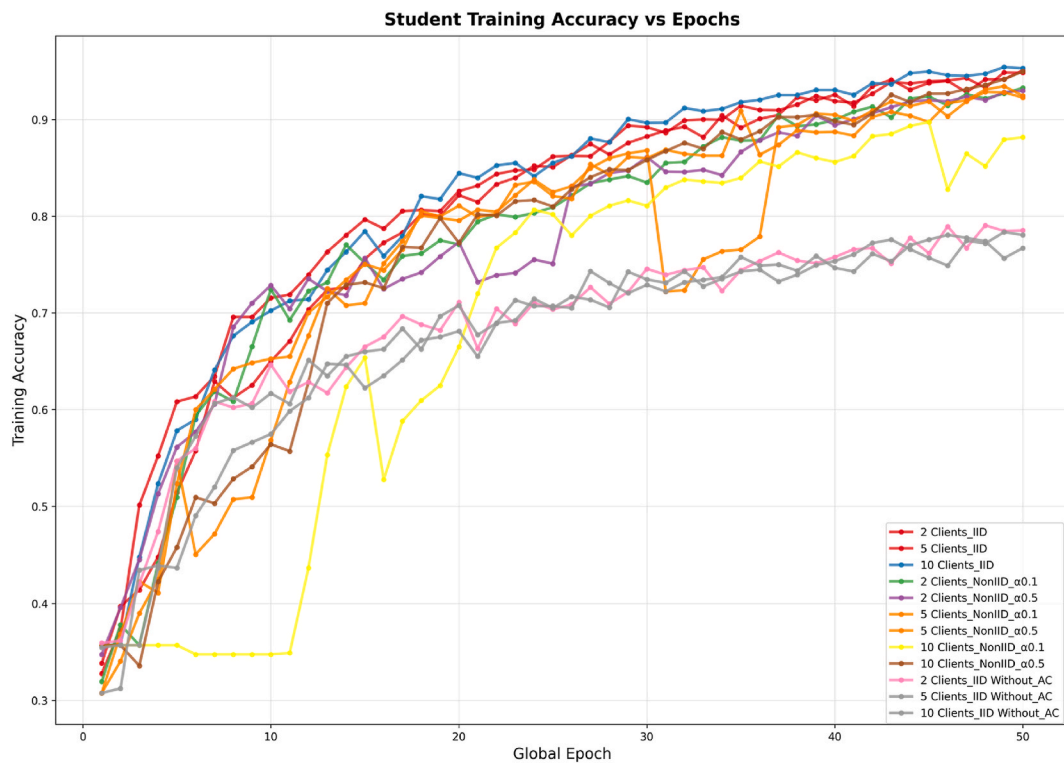


Fig. 8. Training accuracy over epochs in different federation sizes and data distributions.

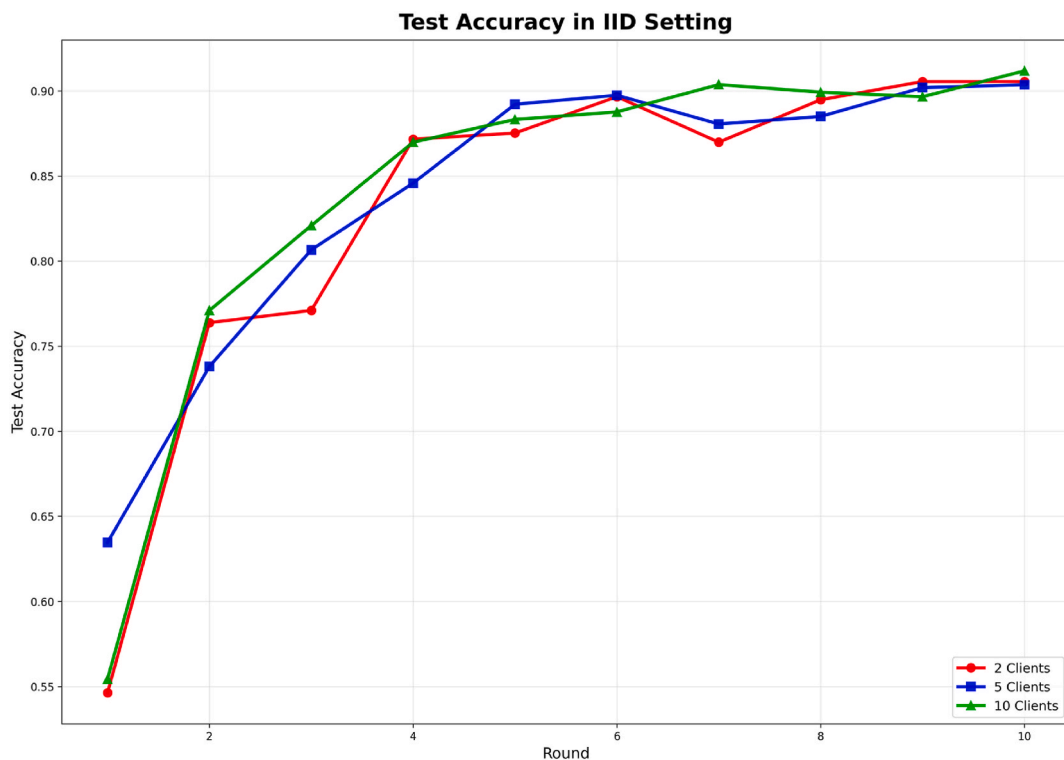


Fig. 9. Test accuracy across communication rounds for 2, 5, and 10 clients in IID data.

setting with $\alpha = 0.1$, these values remained close to 0.90. The removal of AC led to declines across all three metrics, typically to the 0.74 range. The confusion matrices in Fig. 13 illustrate the performance of FedFB with 2, 5, and 10 clients under the non-IID ($\alpha = 0.5$) setting. Across all three configurations, the model achieves balanced recognition of the

trans-cerebellum, trans-thalamic, and trans-ventricular planes. Most errors appear between the trans-cerebellum plane and the other two classes, which reflects the anatomical overlap often encountered in clinical practice. Importantly, the classification patterns remain consistent as the number of clients increases, highlighting the robustness and

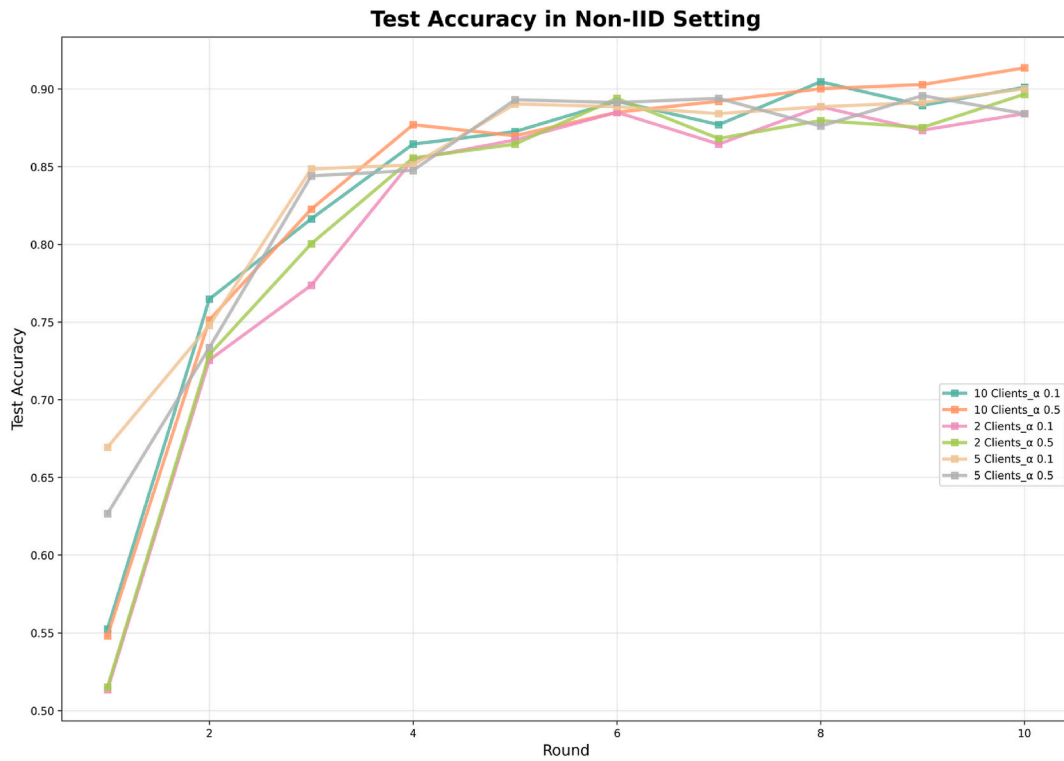


Fig. 10. Test accuracy across communication rounds in non-IID settings with $\alpha = 0.1$ and $\alpha = 0.5$. Even with data imbalance, FedFB converges above 89% after 10 rounds, showing robustness to differences between clients' data.

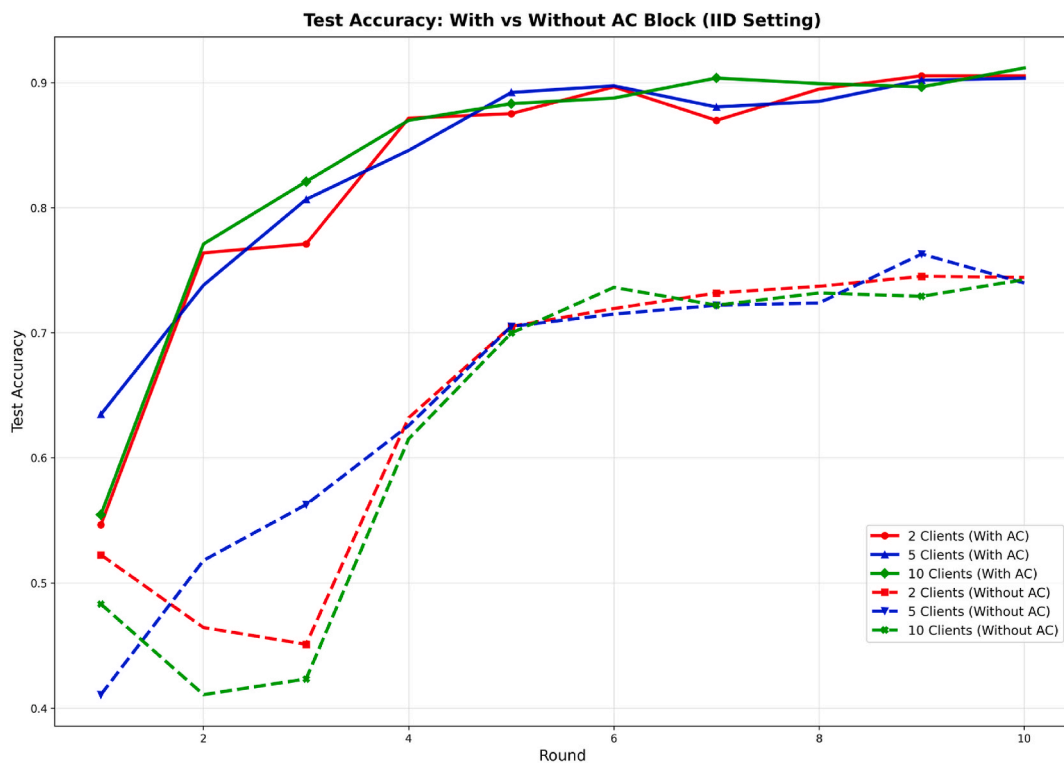


Fig. 11. Test accuracy comparison between models trained with and without the AC module in IID setting.

scalability of the framework.

Fig. 14 compares the validation and test accuracy curves of FedFB with centralized training across 50 epochs. The centralized model converges more rapidly in the initial stages, achieving higher accuracy early

in training. FedFB shows a more gradual improvement but steadily narrows the gap, reaching comparable performance by the later epochs. These results demonstrate that FedFB attains accuracy comparable to centralized training despite the data remaining distributed.

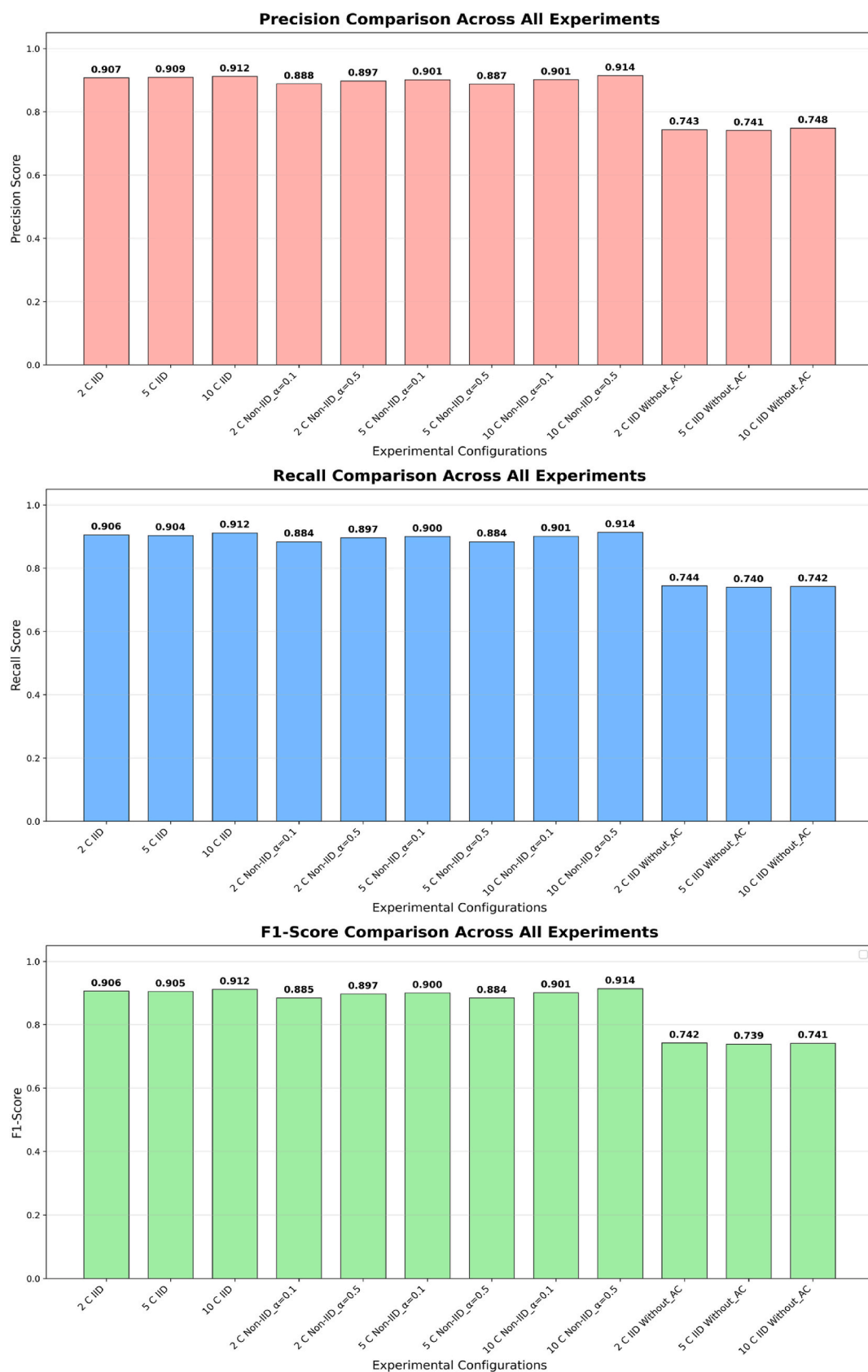


Fig. 12. Precision, Recall and F1-Score comparison for different settings.

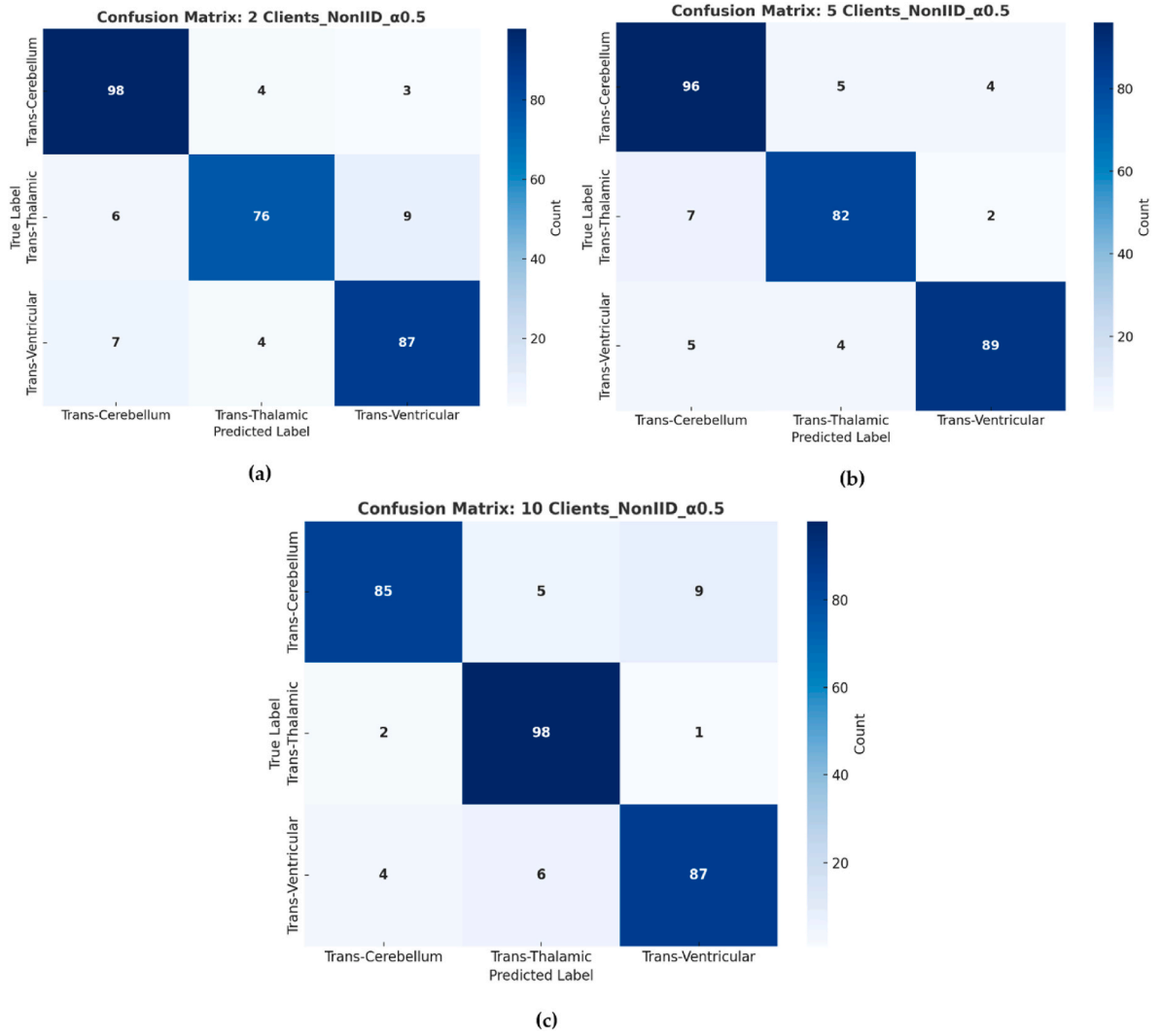


Fig. 13. Confusion matrices of FedFB with 2, 5, and 10 clients (non-IID setting).

The comparative evaluation of FedFB against established FL baselines is summarized in Fig. 15. The selected baselines FedAvg, FedProx, and FedDF represent the most directly comparable approaches for evaluating our framework. FedAvg serves as the foundational parameter aggregation baseline and reflects the behavior of traditional federated averaging. FedProx extends this formulation with proximal regularization to stabilize optimization under heterogeneous data distributions,

providing a robust gradient-based comparison. FedDF constitutes the direct federated distillation baseline, as it shares FedFB objective of producing a single global model through ensemble distillation on auxiliary public data. Other federated distillation methods were excluded because they primarily target personalized FL settings. For example, FedProto [36] and FedMD [37] are designed to support heterogeneous client architectures and optimize client-specific models

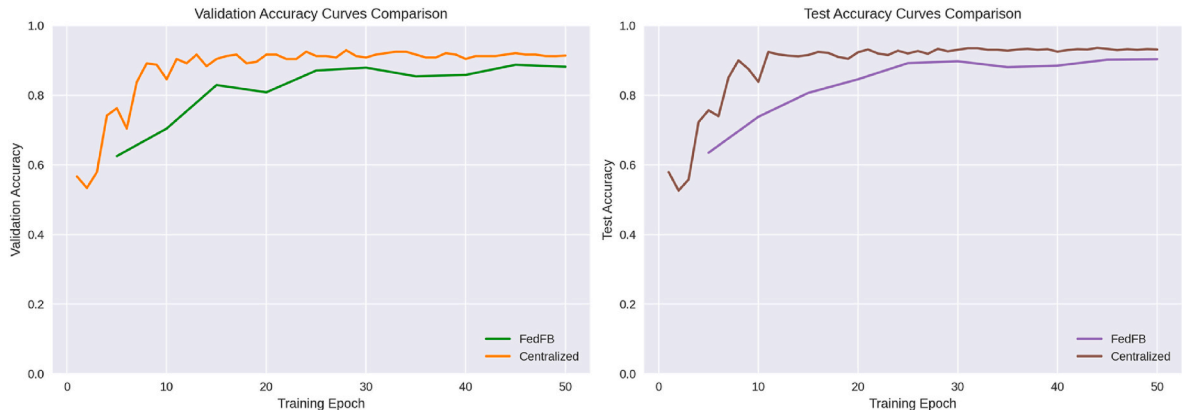


Fig. 14. Validation and test accuracy curves of FedFB compared with centralized training.

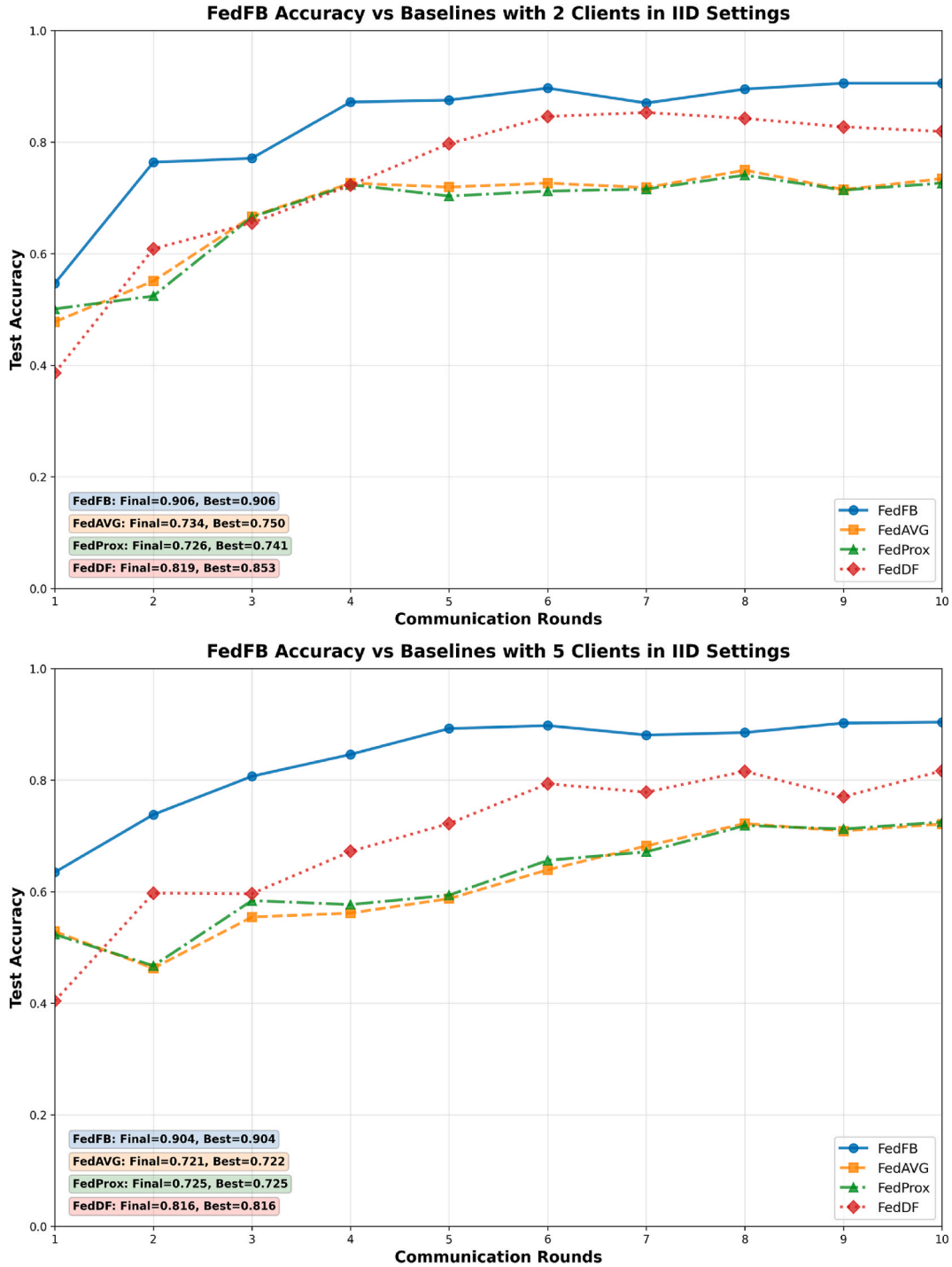


Fig. 15. FedFB vs. baseline methods with 2 and 5 clients in IID setting. FedFB achieves the highest accuracy in all rounds in terms of test accuracy across communication rounds.

rather than learning a single unified global model. FedGKT [38] relies on model partitioning between clients and server and assumes a split architecture design, resulting in a different communication protocol and optimization setting that is not directly comparable to ensemble-based distillation over a shared auxiliary dataset. Generative approaches such as FedGen [39] introduce additional model components and computational overhead, thereby confounding the evaluation of distillation mechanisms themselves. This baseline selection enables a fair evaluation of FedFB mechanism and communication efficiency against both traditional FL and direct federated distillation methods. Across all evaluated IID settings, FedFB consistently outperformed these baselines

in terms of classification accuracy. In the 2-client IID configuration, FedFB achieved a final accuracy of 90%, outperforming FedAvg (73%), FedProx (74%) and FedDF (85%). A similar trend was observed in the 5-client IID setup, where FedFB reached 91%, while the best performing baseline attained only 81%.

In the more challenging non-IID configuration ($\alpha = 0.5$), shown in Fig. 16, the superiority of FedFB remains evident. While all baseline methods exhibited slower convergence and reduced generalization under client heterogeneity, FedFB maintained a smooth and consistent improvement throughout the communication rounds. By the final round, FedFB achieved a test accuracy of 88%, notably surpassing FedDF

(81.6%) and significantly outperforming FedAvg (67%) and FedProx (70%). Unlike the baseline methods, which plateaued early, FedFB continued to improve steadily, reflecting its strong adaptability to non-IID data. Consequently, FedFB demonstrates stable convergence and enhanced generalization even when data distributions among clients are substantially imbalanced.

In terms of communication efficiency, Fig. 17 demonstrates that FedFB requires substantially less data transmission per round compared to conventional FL methods. While FedAvg and FedProx exceed 538 MB with 2 clients and rise above 1.3 GB with 5 clients, and FedDF also incurs several hundred megabytes, FedFB consistently transmits only 15.8 MB. This corresponds to an average reduction of more than 86% across all evaluated scenarios. Such efficiency makes FedFB particularly well suited for deployment in real clinical networks, where limited bandwidth and strict data privacy regulations often restrict the practicality of traditional FL approaches.

4.5. Model interpretability via Grad-CAM

To provide visual interpretability and validate that the student model learned clinically relevant features, we employ Gradient-weighted Class Activation Mapping (Grad-CAM) [40]. Grad-CAM generates saliency maps that highlight the spatial regions most influential to the model's classification decision by computing the gradient of the predicted class score with respect to the feature maps of a target convolutional layer. For the dual-branch student architecture, we extract Grad-CAM visualizations from the last convolutional layer of the backbone as this layer captures high-level semantic features while maintaining sufficient spatial resolution for localization. We selected representative test samples from each of the three anatomical plane classes using the global student model. Images were chosen to represent typical cases with clear anatomical landmarks for each plane.

Fig. 18 illustrates Grad-CAM visualizations generated from the global model trained with five clients under the non-IID ($\alpha = 0.5$) configuration, showing the original fetal brain ultrasound images alongside their corresponding heatmaps. The highlighted regions reveal the spatial focus of the model during classification, indicating that

FedFB consistently attends to key anatomical areas characteristic of each plane. For the trans-ventricular plane, the model's attention concentrates on the bilateral lateral ventricles and the cavum septum pellucidum in the midline region, which are the defining anatomical structures used clinically to identify this plane and measure the ventricular atrium width for detecting ventriculomegaly. For the trans-thalamic plane, activation maps correctly focus on the thalamic nuclei bilaterally and the central cavum septum pellucidum, corresponding to the structures clinicians use to measure the biparietal diameter and assess brain symmetry. For the trans-cerebellum plane, the model appropriately highlights the posterior fossa containing the cerebellum and the cisterna magna, which are critical for evaluating posterior fossa abnormalities such as Dandy-Walker malformation or cerebellar hypoplasia.

Importantly, the heatmaps show minimal activation in image peripheries, maternal tissue boundaries, or acoustic shadows regions that contain artifacts or clinically irrelevant information. This confirms that the model attention is directed toward meaningful structural cues rather than background artifacts or spurious correlations. The Grad-CAM analysis demonstrates that FedFB not only maintains high predictive confidence under data heterogeneity but also provides interpretable visual evidence supporting its decision-making process, with attention patterns that align with established clinical diagnostic criteria. This anatomically grounded interpretability is an important factor for fostering clinical trust and transparency in automated fetal brain assessment, as it allows clinicians to verify that the model's reasoning is consistent with domain expertise.

5. Discussion

The experimental results presented in Section 4 demonstrate that FedFB achieves competitive classification performance while substantially reducing communication costs compared to traditional FL approaches. This section provides a comprehensive discussion of the framework's generalization capability across diverse medical imaging modalities and critically examines its limitations and practical considerations for real-world deployment.

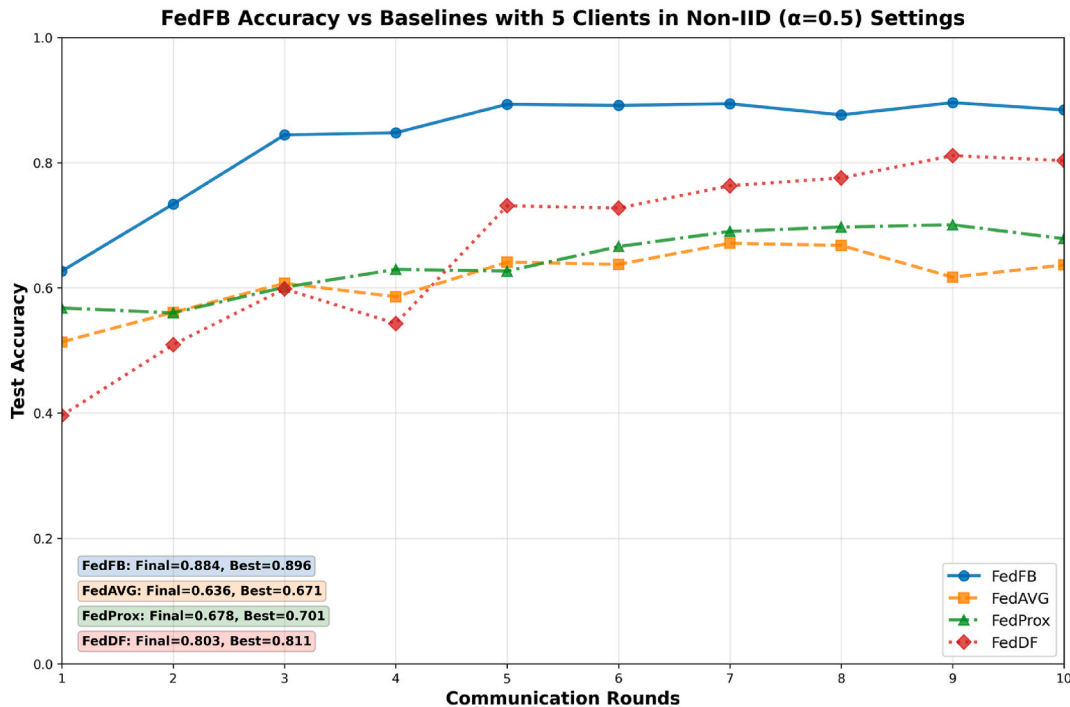


Fig. 16. Comparison of FedFB with baseline FL methods under non-IID ($\alpha = 0.5$) settings using 5 clients, showing FedFB achieving the highest final accuracy despite data heterogeneity.

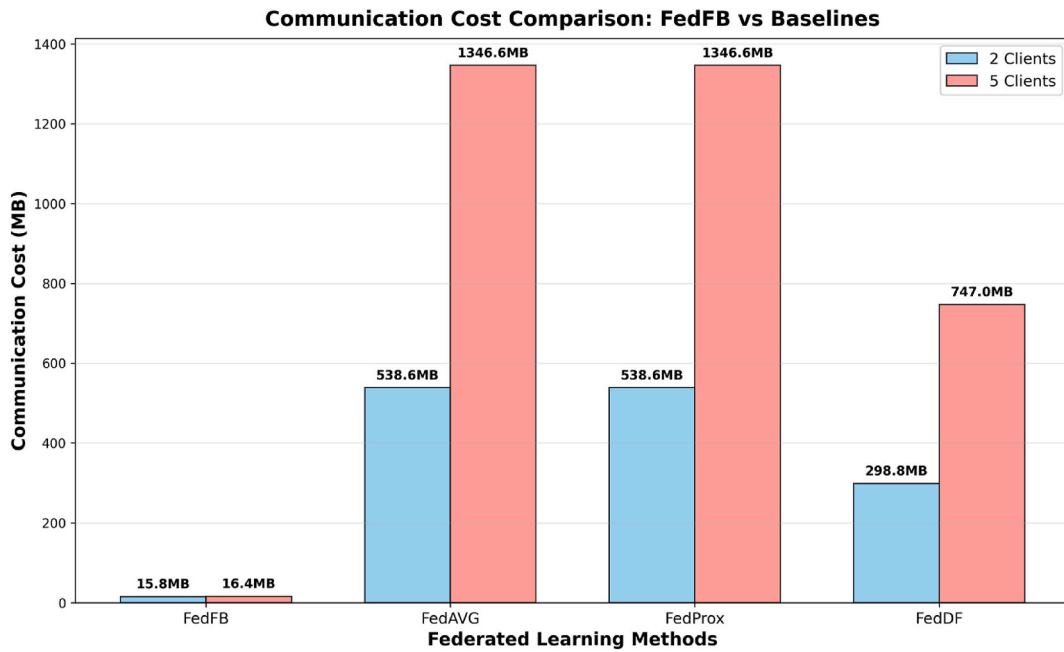


Fig. 17. Comparison of communication costs between FedFB and baselines in IID settings. FedFB achieves much lower cost.

5.1. Validation across medical imaging modalities

The central focus of this work was the development and evaluation of FedFB framework for fetal brain ultrasound plane classification. The proposed method successfully differentiated the three key anatomical planes trans-ventricular, trans-thalamic, and trans-cerebellum even under limited data availability and heterogeneous client distributions. Building on this foundation, FedFB was further extended to brain tumor MRI classification [41] and chest X-ray pneumonia detection [42] to examine its adaptability across diverse medical imaging modalities that similarly face challenges of restricted dataset size and variation in imaging characteristics.

The brain tumor MRI dataset comprised 3060 images across three tumor types: meningioma, glioma, and pituitary. The chest X-ray dataset contained 5800 images categorized as normal, bacterial pneumonia, and viral pneumonia. Both datasets were preprocessed using Contrast Limited Adaptive Histogram Equalization (CLAHE) [43]. This enhancement technique was applied to improve local contrast and emphasize diagnostically relevant regions in each modality, ensuring consistent visual representation and facilitating effective knowledge transfer within the FL framework.

To assess the framework robustness, FedFB was trained under both IID and non-IID ($\alpha = 0.1$ and $\alpha = 0.5$) data distributions using 2, 5, and 10 clients. All experiments were repeated over multiple independent runs with different random seeds, and the reported results correspond to the mean $\pm$ standard deviation to ensure statistical reliability. The comparative test accuracy for the three datasets under the 5-client IID and non-IID ($\alpha = 0.5$) settings are shown in Figs. 19 and 20, respectively. All models exhibited a steady improvement in test accuracy over successive communication rounds. The chest X-ray dataset showed the highest early convergence, followed closely by the brain tumor and fetal brain datasets. This behavior can be attributed not only to the more homogeneous texture and consistent intensity distribution of X-ray images compared with ultrasound images, but also to the larger dataset size available for chest X-ray training. In the non-IID setting illustrated in Fig. 20, performance variations across clients were more pronounced, yet FedFB maintained stable convergence and comparable final accuracy across all datasets. The slight fluctuation observed during the early rounds is typical in non-IID federated scenarios, where client updates

differ significantly in data distribution. Nevertheless, FedFB demonstrated resilience to data imbalance and achieved consistent performance by the final rounds, validating its capability to generalize across varying degrees of heterogeneity.

Across all experiments, FedFB achieved strong accuracy and balanced precision, recall and F1-score performance, while significantly reducing communication overhead by more than 85% compared with conventional parameter sharing methods.

A comparative summary of quantitative results for the fetal brain, brain tumor, and chest X-ray datasets under different configurations is presented in Table 3. All reported metrics correspond to the mean $\pm$ standard deviation over multiple independent runs, providing an assessment of performance stability across different random initializations. These results underscore FedFB suitability for medical applications involving limited and distributed datasets, where privacy, communication efficiency, and data heterogeneity remain key challenges.

5.2. Ablation study

To evaluate the sensitivity of the proposed framework to the size of the auxiliary public dataset, we varied the public fraction $\rho \in \{5\%, 10\%, 20\%, 40\%\}$ under non-IID conditions ($\alpha = 0.5$) across 2, 5, and 10 clients. As shown in Table 4, increasing ρ consistently improves test accuracy for all client configurations. For instance, under 5 clients, performance rises from 61.94% at 5% public data to 89.57% at 40%. A similar monotonic trend is observed for 2 and 10 clients, confirming that richer auxiliary supervision enhances ensemble knowledge transfer during federated distillation, particularly under heterogeneous data distributions.

Importantly, the communication cost increases approximately linearly with both the public fraction ρ and the number of clients, consistent with the theoretical complexity of logit exchange. However, the absolute bandwidth consumption remains low compared to traditional weight-sharing methods, since FedFB transmits only soft predictions rather than full model parameters. These results demonstrate that FedFB effectively leverages additional public data and client diversity to improve accuracy while maintaining strong communication efficiency.

Table 5 presents a focused ablation study to isolate the contribution

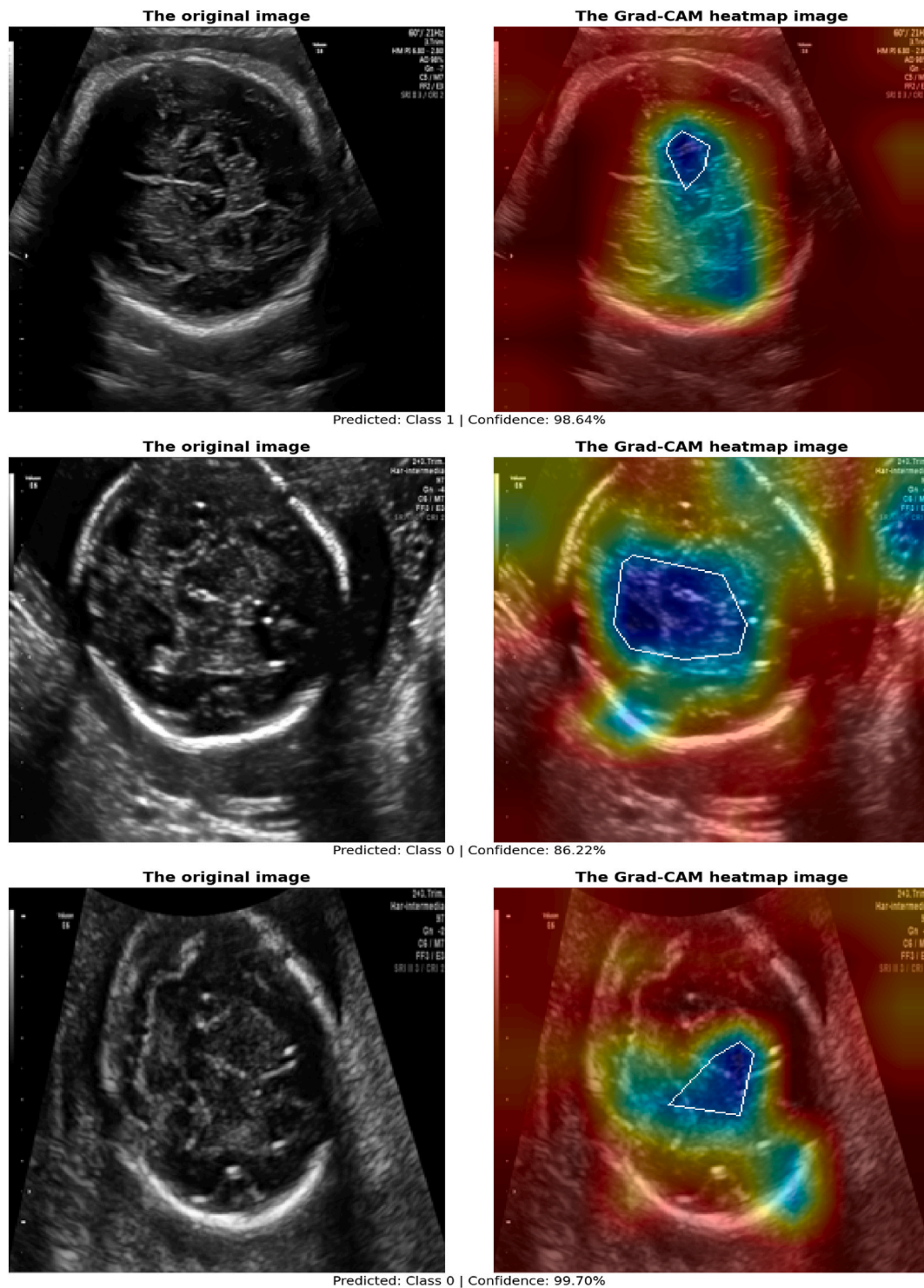


Fig. 18. Grad-CAM visualizations from the global model trained with five clients under the non-IID ($\alpha = 0.5$) setting, showing the original fetal brain ultrasound images alongside their corresponding heatmap overlays. The blue regions represent areas of highest relevance for classification, while the white contours highlight key anatomical structures trans-ventricular, trans-thalamic, and trans-cerebellum that contributed most to the model's decision-making process.

of MLVUM pre-processing within the proposed framework. The baseline Core FedFB corresponds to the pure FKD mechanism without MLVUM enhancement.

The results indicate that the core federated distillation mechanism already achieves strong performance under heterogeneous data conditions, confirming that the primary gains originate from the knowledge transfer framework itself. Incorporating MLVUM yields consistent additional improvements by enhancing local contrast and reducing

ultrasound-specific noise artifacts, thereby improving feature representation prior to distillation. These findings demonstrate that MLVUM provides complementary benefits, while the main performance improvements are driven by the federated distillation strategy.

5.3. Limitations and practical considerations

While FedFB demonstrates strong performance in communication

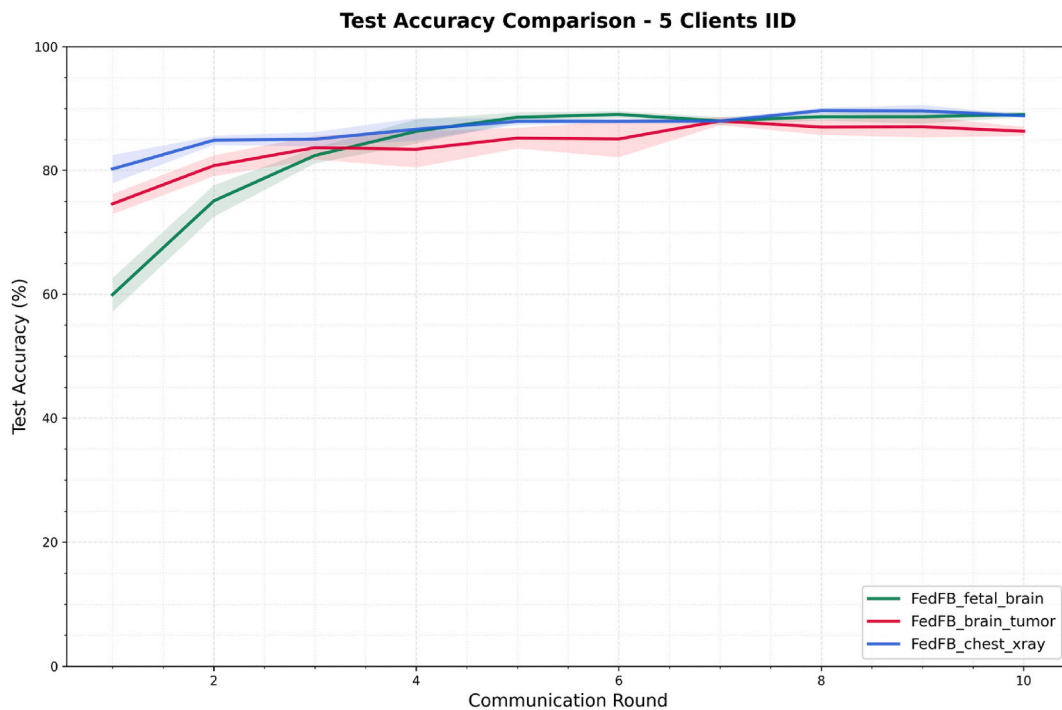


Fig. 19. Test accuracy comparison of FedFB on fetal brain, brain tumor, and chest X-ray datasets under the 5-client IID setting. Results are reported as mean $\pm$ standard deviation over multiple independent runs, with shaded regions indicating variability across communication rounds.

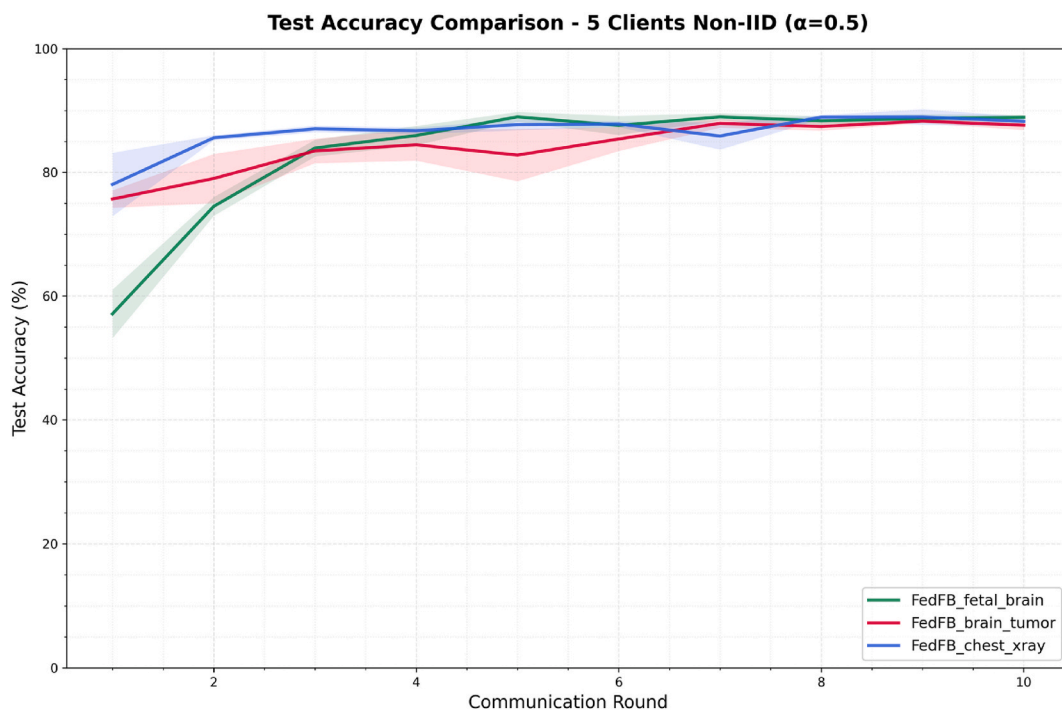


Fig. 20. Test accuracy comparison of FedFB on fetal brain, brain tumor, and chest X-ray datasets under the 5-client non-IID setting ($\alpha = 0.5$). Results are reported as mean $\pm$ standard deviation over multiple independent runs, with shaded regions indicating variability across communication rounds.

efficiency and accuracy across multiple medical imaging tasks, several limitations warrant consideration.

Privacy considerations and logit sharing risks: Regarding privacy guarantees, although FedFB avoids sharing raw patient data or model parameters, the exchange of soft predictions is not entirely immune to privacy risks. Recent studies have shown that logit sharing can be vulnerable to membership inference attacks and model inversion under

adversarial conditions [44], [45]. The temperature parameter in KD provides an implicit privacy-utility tradeoff: higher temperatures smooth the prediction distribution, potentially reducing information leakage but may also diminish distillation quality. To enhance privacy protection in high-risk scenarios, future implementations could integrate differential privacy mechanisms through logit noising or clipping, selective logit sharing strategies, or secure aggregation protocols. It is

Table 3

FedFB performance across datasets and client configurations under IID and non-IID conditions. Accuracy, precision, recall, and F1-score are reported as mean $\pm$ standard deviation over multiple independent runs for the fetal brain, brain tumor, and chest X-ray datasets, demonstrating consistent performance and stable convergence across heterogeneous data distributions.

Dataset		Fetal Brain US			Brain Tumor MRI			Chest X-ray		
		IID	Non-IID		IID	Non-IID		IID	Non-IID	
			$\alpha = 0.1$	$\alpha = 0.5$		$\alpha = 0.1$	$\alpha = 0.5$		$\alpha = 0.1$	$\alpha = 0.5$
2 Clients	Accuracy	89.75	88.09	89.10	88.54	87.60	88.39	88.85	88.79	88.82
		± 0.57	± 0.87	± 0.93	± 0.45	± 1.37	± 0.95	± 1.05	± 0.29	± 1.14
	Precision	89.91	88.54	89.26	88.77	87.71	88.69	88.31	88.85	89.14
		± 0.59	± 0.62	± 0.77	± 0.48	± 1.24	± 0.74	± 0.62	± 0.32	± 0.82
	Recall	89.75	88.09	89.10	88.54	87.60	88.39	88.85	88.79	88.82
		± 0.57	± 0.87	± 0.93	± 0.45	± 1.37	± 0.95	± 1.05	± 0.29	± 1.14
	F1-Score	89.76	88.13	89.13	88.49	87.44	88.29	88.94	88.71	88.89
		± 0.59	± 0.87	± 0.90	± 0.42	± 1.38	± 0.82	± 0.89	± 0.30	± 1.05
5 Clients	Accuracy	89.03	88.87	88.92	88.68	88.18	88.23	89.93	88.57	89.31
		± 1.21	± 0.41	± 0.37	± 1.51	± 0.38	± 0.59	± 0.77	± 0.31	± 0.72
	Precision	89.68	88.09	89.12	88.56	88.20	88.66	90.24	88.95	89.46
		± 0.59	± 0.36	± 0.36	± 1.45	± 0.34	± 0.17	± 0.63	± 0.28	± 0.74
	Recall	89.03	88.87	88.92	88.68	88.18	88.23	89.93	88.57	89.31
		± 1.21	± 0.41	± 0.37	± 1.15	± 0.38	± 0.59	± 0.77	± 0.31	± 0.72
	F1-Score	89.08	88.89	88.92	88.45	88.06	88.26	90.02	88.69	89.33
		± 1.18	± 0.38	± 0.36	± 1.31	± 0.32	± 0.45	± 0.74	± 0.29	± 0.77
10 Clients	Accuracy	89.95	88.75	89.01	88.18	86.73	87.25	89.13	88.44	88.73
		± 1.69	± 0.98	± 0.89	± 1.30	± 0.27	± 1.08	± 0.26	± 0.27	± 0.93
	Precision	89.26	88.84	89.35	88.25	87.27	87.68	89.41	88.76	88.90
		± 1.37	± 0.91	± 0.62	± 1.24	± 0.45	± 0.77	± 0.44	± 0.14	± 0.84
	Recall	88.95	88.75	89.01	88.18	86.73	87.25	89.13	88.44	88.37
		± 1.69	± 0.98	± 0.89	± 1.30	± 0.27	± 1.08	± 0.26	± 0.27	± 0.93
	F1-Score	88.87	88.69	88.81	88.06	86.78	87.26	89.19	88.50	88.48
		± 1.73	± 1.00	± 1.06	± 1.48	± 0.20	± 0.98	± 0.31	± 0.19	± 0.92

Table 4

Impact of varying the public data fraction $\rho \in \{5\%, 10\%, 20\%, 40\%\}$ on test accuracy and communication cost under non-IID ($\alpha = 0.5$) for 2, 5, and 10 clients.

Clients#/Public ρ (%)	Test Accuracy (%)				Communication Cost (MB)			
	5	10	20	40	5	10	20	40
2	63.10%	70.89%	82.60%	89.66%	0.06	0.12	0.23	0.47
5	61.94%	72.10%	82.62%	89.57%	0.11	0.23	0.47	0.94
10	62.66%	72.37%	81.90%	91.35%	0.21	0.43	0.86	1.72

Table 5

Ablation analysis of FedFB under 5 clients, non-IID ($\alpha = 0.5$) and ($\rho = 0.4$).

Method	MLVUM	Test Accuracy (%)
Core FedFB	×	87.29
FedFB	✓	89.57

important to clarify that FedFB's current design reduces privacy exposure by avoiding direct data and weight sharing, but does not offer formal privacy guarantees against sophisticated inference attacks without additional protective mechanisms.

Computational Requirements and Domain Generalization: Computational requirements on client devices merit attention. Training ensemble teacher models (ResNet, VGG16, DenseNet) demands substantial local resources, which may pose challenges in resource-constrained clinical environments. While the student model itself is lightweight and efficient for inference, the initial teacher training phase requires careful resource allocation. Additionally, the current evaluation focuses exclusively on medical imaging modalities (ultrasound, MRI, X-ray) generalization to non-medical domains or significantly different data types has not been assessed.

Auxiliary Dataset Dependence and Scalability: The framework relies on the availability and representativeness of an auxiliary public dataset for knowledge transfer. In this work, relatively small auxiliary datasets up to 450 samples in our experiment were sufficient to achieve effective distillation. However, the representativeness of this auxiliary

data relative to the private client distributions can significantly impact student model performance, particularly when domain gaps are substantial. Furthermore, while FedFB has been validated with up to 10 clients, scalability to larger federated networks (e.g., hundreds of clients) remains to be empirically verified. Although communication costs scale linearly with the number of clients in theory, practical considerations such as aggregation complexity, synchronization overhead, and the potential for client dropout or network heterogeneity in real-world deployments require further investigation. Asynchronous training schemes and robust aggregation methods could be explored to address intermittent client participation.

Practical Construction of the Auxiliary Dataset: The practical feasibility of constructing the auxiliary dataset in clinical settings requires consideration. Although FedFB significantly reduces communication cost and privacy risks compared to traditional FL, the need for an auxiliary dataset introduces a limited shared-data requirement. A common approach is for each participating institution to contribute a small, de-identified subset of images prior to federated training. For example, five hospitals could each share 200 anonymized images (1000 total), typically representing less than 10% of their data. This one-time transfer would occur under institutional review board approval and data use agreements, with full metadata removal and anonymization. Alternatively, existing public repositories such as the dataset used in this study [31] can serve as auxiliary data, eliminating inter-institutional transfers entirely. From a privacy standpoint, sharing a small, anonymized subset poses substantially lower exposure than centralized training, as most institutional data remain local. Institutions must balance this limited

data sharing against FedFB's advantages, including over 85% communication reduction and no model weight exchange. For highly sensitive scenarios, differential privacy or synthetic data generation could further mitigate risks, potentially at some performance cost. Ultimately, feasibility depends on governance policies, regulatory constraints, and public dataset availability, though many research collaborations may find this trade-off acceptable given the framework's communication efficiency and reduced data exposure.

6. Conclusion and future work

This paper presents FedFB, a federated KD framework developed for automated classification of fetal brain ultrasound images into the trans-ventricular, trans-thalamic, and trans-cerebellum planes. The proposed global model employs a dual-branch student architecture, in which a lightweight backbone is complemented by an auxiliary attention-convolution branch to enhance feature representation without increasing computational complexity. To further improve image quality before training, Maximum Local Variation-based Unsharp Masking was applied to suppress speckle noise and enhance contrast, enabling more stable feature extraction for both teacher and student models. Unlike conventional FL strategies that rely on test set metrics for tuning, FedFB adapts training and model selection based on validation performance, eliminating test leakage and ensuring methodological rigor. To assess its generalization capability, FedFB was also implemented on brain tumor MRI and chest X-ray datasets, confirming consistent performance across distinct imaging modalities and limited size datasets affected by heterogeneity. In a 10-client non-IID configuration ($\alpha = 0.5$), FedFB achieved approximately 91% accuracy, precision, recall, and F1-score, while reducing communication overhead by more than 85% compared with traditional FL frameworks. These findings demonstrate that FedFB delivers both robust accuracy and communication efficiency, making it highly suitable for privacy sensitive and data limited clinical environments. Future research will extend FedFB beyond standard plane classification toward multi-task fetal brain analysis, including segmentation of critical anatomical structures such as the ventricles and cerebellum, and early detection of developmental abnormalities like ventriculomegaly and Dandy-Walker malformation. Further directions include exploring adaptive teacher-student architectures and personalized federated distillation to improve interpretability and clinical reliability across diverse healthcare settings.

Informed consent and patient details

The authors declare that this report does not contain any **personal information** that could lead to the identification of the patient(s) and/or volunteers.

The authors declare that they obtained a written **informed consent** from the patients and/or volunteers included in the article and that this report does not contain any **personal information** that could lead to their identification.

The authors declare that the work described does not involve patients or volunteers.

Ethical statement

This study did not involve human participants, animal subjects, or the use of identifiable or sensitive data.

Authors' contribution

HS contributed to Conceptualization, Methodology, Investigation, Experiments and Writing the Original Draft. JFP contributed to Supervision, Methodology, Review & Editing. SG contributed to Review & Editing. NC, CZ, and AN contributed to Validation, Review & Editing. All authors read and approved the final manuscript.

Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE)

Criteria for authorship

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship. Individual author contributions are as follows: HS contributed to Conceptualization, Methodology, Investigation, Experiments and Writing the Original Draft. JFP contributed to Supervision, Methodology, Review & Editing. SG contributed to Review & Editing. NC, CZ, and AN contributed to Validation, Review & Editing. All authors read and approved the final manuscript.

Funding

This work did not receive any **grant** from funding agencies in the public, commercial, or not-for-profit sectors.

This work has been **supported** by: V5.0 – 27/08/19 – For use with journals published by Elsevier Masson SAS, France.

Disclosure of interest

The authors declare that they have no known **competing financial or personal relationships** that could be viewed as influencing the work reported in this paper.

The authors declare the **following financial or personal relationships** that could be viewed as influencing the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.neuri.2026.100272>.

References

- [1] G. Litjens, T. Kooi, B. Bejnordi, A. Setio, FC-M image, 2017 undefined. A survey on deep learning in medical image analysis, in: ElsevierG. Litjens, T. Kooi, B. E. Bejnordi, A.A.A. Setio, F. Ciompi, M. Ghahfarian (Eds.), JA Van Der LaakMedical Image Analysis, Elsevier n.d, 2017.
- [2] G. Kaissis, M. Makowski, DR-NM, 2020 undefined. Secure, privacy-preserving and federated machine learning in medical imaging, in: Kaissis NatureComGA, M. R. Makowski, D. Rückert (Eds.), RF BrarenNature Machine Intelligence, natureCom n.d, 2020.
- [3] D. Shen, G. Wu, Suk H. II, Deep learning in medical image analysis, AnnualreviewsOrg 19 (2017) 221–248, <https://doi.org/10.1146/ANNUREV-BIOENG-071516-044442>.
- [4] J. Wen, Z. Zhang, Y. Lan, Z. Cui, J. Cai, W. Zhang, A survey on federated learning: challenges and applications, International Journal of Machine Learning and Cybernetics 14 (2) (2022) 513–535, <https://doi.org/10.1007/S13042-022-01647-Y>, 2022;14.
- [5] J. Gou, B. Yu, S.J. Maybank, D. Tao, Knowledge distillation: a Survey, Int. J. Comput. Vis. 129 (2021) 1789–1819, <https://doi.org/10.1007/S11263-021-01453-Z/METRICS>.
- [6] H. Salman, C. Zaki, N. Charara, S. Guehis, J.F. Pradat-Peyre, A. Nasser, Knowledge distillation in federated learning: a comprehensive survey, Discover Computing 28 (2025) 1–40, <https://doi.org/10.1007/S10791-025-09657-4/TABLES/10>.
- [7] H. Chen, P. Bai, S. Yang, M. Jia, H. Tian, C. Jz-J of, et al., Short-term and long-term outcomes of fetal ventriculomegaly beyond gestational 37 weeks: a retrospective cohort study, in: MdpiComH. Chen, P. Bai, S. Yang, M. Jia, H. Tian, J. Zou (Eds.), X XiaoJournal of Clinical Medicine, mdpiCom n.d, 2023.
- [8] E. Snider, S. Hernandez-Torres, Reports EB-S, 2022 undefined. An image classification deep-learning algorithm for shrapnel detection from ultrasound images, in: NatureComEJ. Snider, S.I. Hernandez-Torres (Eds.), EN BoiceScientific Reports, vol. 12, natureCom 123AD, 2022, p. 8427, <https://doi.org/10.1038/s41598-022-12367-2>.
- [9] R. Qu, G. Xu, C. Ding, W. Jia, M.S.-I. Access, Undefined. Standard plane identification in fetal brain ultrasound scans using a differential convolutional neural network, in: IeeeexploreIeeeOrgR. Qu, G. Xu, C. Ding, W. Jia (Eds.), M SunIEEE Access, 2020•IeeeexploreIeeeOrg n.d, 2020.
- [10] Y. Jalali, M. Fateh, M. Rezvani, V. Abolghasemi, M.A.- Sensors, 2021 undefined. ResBCDU-Net: a deep learning framework for lung CT image segmentation, in:

- MdpiComY. Jalali, M. Fateh, M. Rezvani, V. Abolghasemi (Eds.), MH AnisiSensors, mdpiCom n.d, 2021.
- [11] T. Krishna, Applications PK-ES with, 2024 undefined. Standard fetal ultrasound plane classification based on stacked ensemble of deep learning models, in: ElsevierTB. Krishna, P. KokilExpert (Eds.), Systems with Applications, Elsevier n.d, 2024.
 - [12] E. Jain, P. Kaushik, V. Kukreja, Dogra A. Sakshi, B. Goyal, Fetal diagnostics using vision transformer for enhanced health and severity prediction in ultrasound imaging, in: BenthamdirectComE. Jain, P. Kaushik, V. Kukreja, A Dogra Sakshi (Eds.), B GoyalCurrent Medical Imaging, vol. 21, 2025•benthamdirectCom, 2025, <https://doi.org/10.2174/0115734056360199250227053012>.
 - [13] L. Zhao, K. Li, B. Pu, J. Chen, S. Li, X. Liao, An ultrasound standard plane detection model of fetal head based on multi-task learning and hybrid knowledge graph, Future Gener. Comput. Syst. 135 (2022) 234–243, <https://doi.org/10.1016/J.FUTURE.2022.04.011>.
 - [14] D. Vetrivelvi, R. Thenmozhi, Prenatal diagnostics using deep learning: a dual approach to plane localization and cerebellum segmentation in ultrasound images, J. Clin. Ultrasound 53 (2025) 721–735, <https://doi.org/10.1002/JCU.23924>.
 - [15] R.F. Gusa, R. Hidayat, H.A. Nugroho, Speckle noise reduction in ultrasound images using various denoising techniques. Proceedings - International Seminar on Intelligent Technology and Its Applications, ISITIA, 2024, pp. 338–342, <https://doi.org/10.1109/ISITIA63062.2024.10667756>.
 - [16] C. Di Vece, B. Dromey, F. Vasconcelos, A.L. David, D. Peebles, D. Stoyanov, Deep learning-based plane pose regression in obstetric ultrasound, Int. J. Comput. Assist. Radiol. Surg. 17 (2022) 833–839, <https://doi.org/10.1007/S11548-022-02609-Z/FIGURES/5>.
 - [17] C. Di Vece, A. Cirigliano, M Le Lous, R. Napolitano, A.L. David, D. Peebles, et al., Measuring Proximity to Standard Planes During Fetal Brain Ultrasound Scanning, 2024.
 - [18] T. Yu, P.H. Tsui, D. Leonov, S. Wu, G. Bin, Z. Zhou, LPC-SonoNet: a lightweight network based on SonoNet and light Pyramid convolution for fetal ultrasound standard plane detection, Sensors 24 (2024), <https://doi.org/10.3390/S24237510>, 7510 2024;24:7510.
 - [19] A. Sivasubramanian, D. Sasidharan, V. Sowmya, V. Ravi, Efficient feature extraction using light-weight CNN attention-based deep learning architectures for ultrasound fetal plane classification, Phys Eng Sci Med (2025) 1–15, <https://doi.org/10.1007/S13246-025-01566-6/METRICS>.
 - [20] AC.Branch, MobileNet: an effective hybrid branch network for plane classification in fetus brain ultrasound image, Biomed. Signal Process Control 112 (2026) 108586, <https://doi.org/10.1016/J.BSPC.2025.108586>.
 - [21] M.N. Hasan, A.B.M. Aowlad Hossain, Fetal brain planes classification using deep ensemble transfer learning from U-Net segmented fetal neurosonography images, Int. J. Image Graph. Signal Process. 16 (2024) 74–86, <https://doi.org/10.5815/IJIGSP.2024.04.06>.
 - [22] A. Montero, E. Bonet-Carne, X.B.-A.- Sensors, 2021 undefined. Generative adversarial networks to improve fetal brain fine-grained plane classification, in: MdpiComA. Montero, E. Bonet-Carne (Eds.), XP Burgos-ArtizzuSensors, mdpiCom n.d, 2021.
 - [23] J. Dapuerto, L. Zini, F. Odone, Knowledge distillation for efficient standard scanplane detection of fetal ultrasound, Med. Biol. Eng. Comput. 62 (2024) 73–82, <https://doi.org/10.1007/S11517-023-02881-4/FIGURES/4>.
 - [24] F. Rauf, M.A. Khan, A.K. Bashir, K. Jabeen, A. Hamza, A.I. Alzahrani, et al., Automated deep bottleneck residual 82-layered architecture with Bayesian optimization for the classification of brain and common maternal fetal ultrasound planes, Front. Med. 10 (2023), <https://doi.org/10.3389/FMED.2023.1330218>.
 - [25] M. Kang, P. Chikontwe, S. Kim, K.H. Jin, E. Adeli, K.M. Pohl, et al., One-Shot federated learning on medical data using knowledge distillation with image synthesis and client model adaptation, in: Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 14221 LNCS, 2023, pp. 521–531, https://doi.org/10.1007/978-3-031-43895-0_49.
 - [26] Y. Ma, H. Zhang, Q. Yang, G. Luo, Y. Zhu, A New One-Shot Federated Learning Framework for Medical Imaging Classification with Feature-Guided Rectified Flow and Knowledge Distillation, 2025.
 - [27] Y. Jiang, C.M. Feng, J. Ren, J. Wei, Z. Zhang, Y. Hu, et al., From pretraining to privacy: federated ultrasound foundation model with self-supervised learning, npj Digit. Med. (8) (2025 8:1 2025) 714, <https://doi.org/10.1038/s41746-025-02085-0>.
 - [28] J. Shao, F. Wu, J. Zhang, Selective knowledge sharing for privacy-preserving federated distillation without a good teacher, Nat. Commun. 15 (1) (2024) 1–11, <https://doi.org/10.1038/s41467-023-44383-9>, 2024;15.
 - [29] A. Howard, M. Sandler, B. Chen, W. Wang, L.C. Chen, M. Tan, et al., Searching for MobileNetV3. Proceedings of the IEEE International Conference on Computer Vision, 2019, pp. 1314–1324, <https://doi.org/10.1109/ICCV.2019.00140>.
 - [30] E. Parcham, M.F.-N. Networks, Undefined. HybridBranchNet: a novel structure for branch hybrid convolutional neural networks architecture, in: ElsevierE. Parcham, M. Fateh (Eds.), Neural Networks, 2023•Elsevier n.d, 2023.
 - [31] X.P. Burgos-Artizzu, D. Coronado-Gutiérrez, B. Valenzuela-Alcaraz, E. Bonet-Carne, E. Eixarch, F. Crispí, et al., Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes, in: NatureComXP. Burgos-Artizzu, D. Coronado-Gutiérrez, B. Valenzuela-Alcaraz, E. Bonet-Carne (Eds.), E EixarchScientific Reports, vol. 10, 2020•natureCom, 2020 10200, <https://doi.org/10.1038/s41598-020-67076-5>.
 - [32] S. Lakshmi, D. Rajasekar, A. Purayath, V. Maik, M. Lakshmanan, M. Sivaprakasam, Enhancement of fetal brain ultrasound image via anatomical ROI extraction. Proceedings of 2024 International Conference on Brain Computer Interface and Healthcare Technologies, ICON-BCIHT, 2024 2024, pp. 195–200, <https://doi.org/10.1109/ICON-BCIHT63907.2024.10882410>.
 - [33] B. McMahan, E. Moore, DR-A intelligence, 2017 undefined. Communication-efficient learning of deep networks from decentralized data, in: ProceedingsMlrPressB. McMahan, E. Moore, D. Ramage, S. Hampson (Eds.), BA Y Arcasartificial Intelligence and Statistics, proceedingsMlrPress n.d, 2017.
 - [34] T. Li, A.K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, V. Smith, Federated optimization in heterogeneous networks, in: ProceedingsMlrsysOrgT. Li, A.K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar (Eds.), V SmithProceedings of Machine Learning and Systems, 2020•proceedingsMlrsysOrg, 2020.
 - [35] T. Lin, L. Kong, S. Stich, M.J.-A. neural, 2020 undefined. Ensemble distillation for robust model fusion in federated learning, in: ProceedingsNeuripsCcT. Lin, L. Kong, S.U. Stich, M. JaggiAdvances (Eds.), Neural Information Processing Systems, proceedingsNeuripsCc n.d, 2020.
 - [36] Y. Tan, G. Long, L. Liu, T. Zhou, Q. Lu, J. Jiang, et al., FedProto: Federated prototype learning across heterogeneous clients, Proc. AAAI Conf. Artif. Intell. 36 (2022) 8432–8440, <https://doi.org/10.1609/aaai.v36i8.20819>.
 - [37] D. Li, J. Wang, FedMD: heterogenous federated learning via model distillation. <https://doi.org/10.48550/arXiv.1910.03581>, 2019.
 - [38] C. He, M. Annavaram, S. Avestimehr, Group knowledge transfer: Federated learning of large CNNs at the edge, Adv. Neural Inf. Process. Syst. 33 (2020) 14068–14080.
 - [39] Z. Zhu, J. Hong, Machine JZ-1 conference on, 2021 undefined. Data-free knowledge distillation for heterogeneous federated learning, in: ProceedingsMlrPressZ. Zhu, J. Hong (Eds.), J ZhouInternational Conference on Machine Learning, proceedingsMlrPress n.d, 2021.
 - [40] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: visual explanations from deep networks via gradient-based localization, in: OpenaccessThecvfComRR. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh (Eds.), D BatraProceedings of the IEEE International Conference on Computer, 128, openaccessThecvfCom n.d, 2017, pp. 336–359, <https://doi.org/10.1007/s11263-019-01228-7>.
 - [41] J. Cheng, Brain Tumor Dataset. V1, Science Data Bank, 2022, <https://doi.org/10.57760/sciencedb.06290>.
 - [42] D. Kermany, K. Zhang, M. Goldbaum, Labeled optical coherence tomography (oct) and chest x-ray images for classification, 2, <https://doi.org/10.17632/RSCBJBR9SJ.2>, 2018.
 - [43] R. Sharma, A. Kamra, A review on CLAHE based enhancement techniques. Proceedings of International Conference on Contemporary Computing and Informatics, IC3I 2023, 2023, pp. 321–325, <https://doi.org/10.1109/IC3I59117.2023.10397722>.
 - [44] Z. Yang, Y. Zhao, J. Zhang, Fd-leaks: membership inference attacks against federated distillation learning, in: SpringerZ. Yang, Y. Zhao (Eds.), J ZhangAsia-Pacific Web (Apweb) and Web-Age Information Management (WAIM) Joint, Springer 2023;13423 LNCS, 2022, pp. 364–378, https://doi.org/10.1007/978-3-031-25201-3_28.
 - [45] H. Takahashi, J. Liu, YL the I conference on, 2023 undefined. Breaching FedMD: image recovery via paired-logits inversion attack, in: OpenaccessThecvfComH. Takahashi, J. Liu (Eds.), Y LiuProceedings of the IEEE/CVF Conference on Computer Vision and, openaccessThecvfCom n.d, 2023.